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Quivers, superconformal field theories and higher symmetries

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Abstract
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Quantum field theory (QFT) currently provides our best description of the natural world. Over the past century, it has been able to accurately reproduce many experimental results to an astonishing degree of accuracy and predict the existence of new particles. Despite this, QFT suffers from several mathematical problems. One such problem is the fact that it is often only possible to obtain perturbative results in a weak coupling regime rather than exact results. However, if one constrains the system by introducing supersymmetry, exact results become more tractable. In this thesis we aim to explore various non-perturbative features of QFT by working with theories with eight supercharges.

Much of the material of this thesis aims to understand QFTs from the point of view of BPS quivers-- graphical objects that govern the worldlines of BPS states. By understanding the geometric engineering of these objects, we then use them to extract the higher symmetries of many theories. We also relate disconnected quivers in the orbifold case to the decomposition of two dimensional sigma models.

A second problem tackled in this thesis is that of the classification problem of four dimensional conformal theories with eight supercharges. More specifically, we define a quantity which allows us to single out theories whose Donagi-Witten integrable system simplifies tremendously. The simplifications make the classification problem more tractable for this subset of theories and we initiate the study of them by presenting two new putative rank-2 theories.

Keywords: Quantum field theory, Supersymmetry, Conformal field theory, Quivers, Seiberg-Witten theory, Geometric engineering, Higher symmetries

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List of papers

This thesis is based on the following papers, which are referred to in the text by their Roman numerals.

I S.S. Hosseini & R. Moscrop, *Maruyoshi-Song flows and defect groups of $D_p^b(G)$ theories*, JHEP **10** (2021), [arXiv:2106.03878].

II M. Del Zotto, J.J. Heckman, S.N. Meynet, R. Moscrop & H.Y. Zhang, *Higher symmetries of 5d orbifold SCFTs*, Phys. Rev. D **106** (2022), [arXiv:2201.08372].

III S.N. Meynet & R. Moscrop, *McKay quivers and decomposition*, submitted to Lett. Math. Phys. (2022), [arXiv:2208.07884].

IV S. Cecotti, M. Del Zotto, M. Martone & R. Moscrop, *The Characteristic Dimension of Four-dimensional $\mathcal{N} = 2$ SCFTs*, Comm. Math. Phys. (2023), [arXiv:2108.10884].

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Contents

1. Introduction

Quantum field theory is by far our best description of the natural world. It can reproduce many non-trivial physical results to an astonishing degree of accuracy and predict the existence of new particles. Despite its success, it suffers from many mathematical maladies which raise several questions. These questions are often as fundamental as asking what the definition of a quantum field theory really is, to which we can only give partial answers or perhaps none at all.

One unfortunate truth of quantum field theory is the fact that it is often only possible to obtain results perturbatively in a weak coupling regime. However, if one constrains a system by imposing a sufficient amount of symmetry, then non-perturbative results become more tractable. In particular, there has been great success in doing this in *supersymmetric* systems. Supersymmetry is a symmetry between two types of fields: fermionic and bosonic. These two types of field differ due to the remarkable spin-statistics theorem [1, 2] which relates the spin of a particle to whether their wave functions are symmetric or anti-symmetric under the exchange of particles. The observable world, at least at our energy scale, is not supersymmetric, but the simplifications and beautiful mathematics it introduces makes supersymmetric field theories an ideal laboratory to explore both new physics and mathematics.

The introduction of supersymmetry to a system opens up a whole range of new tools which, among other things, gives us access to results in the strong coupling regime and provides methods for studying systems without a conventional Lagrangian description. Two particularly important concepts for the work presented in this thesis are that of *Seiberg-Witten geometry* [3, 4] and *BPS quivers* [5].

The seminal work of Seiberg and Witten [3, 4] provides an exact solution for the dynamics of a four dimensional $\mathcal{N} = 2$ theory in terms of Riemann surfaces endowed with a meromorphic one form. From these geometric considerations, one can understand many non-trivial features of the theory, including the spectrum of BPS states. Such states are the lowest mass excitations in each charge superselection sector for the Hilbert space of a theory and can provide invaluable information about said theory. A particularly interesting insight given to us by the Seiberg-Witten set up is the fact that one can construct many consistent theories that possess no Lagrangian [6]. While these theories were once thought to be impossible, they now form an important part of the landscape of *superconformal field theories* (SCFTs).

SCFTs possess an enlarged spacetime symmetry group which incorporates a scale invariance. Theories of this type appear as fixed points of RG flows and have myriad uses in modern theoretical physics. For example, they occupy a central role in the famed AdS/CFT correspondence [7], realise interesting gauge theory dualities through S-duality [8] and allow for exact computation of correlators and indices due to their deep mathematical structure [9].

While SCFTs are incredibly useful, they are also somewhat sparse in the landscape of all SQFTs. It is therefore hopeful that the Seiberg-Witten methodology can help classify, or at least identify a large subset of, such theories. This has been studied extensively with many positive results [10–16]. For example, all possible rank-1 SCFTs have been classified through low energy considerations alone together with the Kodaira classification of elliptic fibres [17]. The rank-2 case, while still open, is becoming increasingly well explored. Contributing to this classification scheme is one of the main focusses of the research presented in this thesis, with paper IV doing so by initiating a study of *isotrivial theories*. Such theories have remarkable simplifications at the level of the Seiberg-Witten geometry, giving hope of classifying such theories at arbitrary rank.

Another useful tool for our purposes is that of a BPS quiver. BPS states of four dimensional $\mathcal{N} = 2$ theories preserve four supercharges along their worldlines. Hence, their worldlines are governed by a supersymmetric quantum mechanics with four conserved supercharges. In many cases these SQMs are explicitly known and are described by a quiver gauge theory. Such a quiver gauge theory is referred to as the BPS quiver of the theory. Viewed from the point of view of the four dimensional theory, the BPS quiver is simply a directed graph determined from the charge lattice of the theory which, in turn, encodes this quiver quantum mechanics [5]. Stated in this way, a BPS quiver is nothing more than a graphical tool. However, using the mathematical machinery of quiver representations, we can calculate explicit BPS chambers of the theory. From such information one can then obtain intricate information about the theory, such as the Schur index, flavour algebra and Coulomb branch scaling dimensions [18–20]. This is particularly useful when one considers BPS quivers from the perspective of *geometric engineering*. By taking this point of view, the BPS quiver is encoding branes wrapping volume minimising cycles in the internal geometry of type IIB string theory. This therefore allows us to form the BPS quiver of a theory by simply knowing the homology of the internal geometry together with how such cycles intersect. Doing so, thanks to geometric engineering, we can compute explicitly the BPS quivers for many theories without a conventional Lagrangian description and obtain an object which encodes the BPS spectrum of a theory without using knowledge of a single BPS chamber explicitly!

The geometric engineering perspective also relates the BPS quiver to the notion of *higher form symmetries* [21]. Higher form symmetries are the symmetries of extended objects in a theory and form a part of the paradigm shift that is 'global categorical symmetries'. This paradigm rephrases the guiding principle of symmetry in a more general way, allowing for new types of symmetry in physics. A major aim of the work presented in this thesis is to use non-perturbative data to infer information about the higher symmetries of a theory. In particular, we use BPS quivers in papers I and II to study the one form symmetries of various four and five dimensional theories. Furthermore, we use an extreme consequence of higher form symmetries to study *McKay quivers*, an object which summarises the representation ring of a finite group. By doing this, we see that a disconnected McKay quiver properly reflects the notion of decomposition in the context of two-dimensional sigma models (paper III).

Outline. This thesis is split into three parts followed by papers I–IV. The aim of the three parts is to simply give an overview of preliminary material required for papers I–IV. We aim to be pedagogical and reference useful detailed reviews for the interested reader throughout.

Part I gives a preliminary survey of four dimensional $\mathcal{N} = 2$ theories and their low energy properties. Particular emphasis will be put on BPS quivers and the geometric structure of the moduli space as to prepare for papers I, II and IV. Part II will recontexualise many features of both four dimensional and five dimensional theories from the point of view of string theory and geometric engineering. Finally, part III will detail the recent the redevelopment of symmetries into what is now known as generalised symmetries or global categorical symmetries. We will highlight an extreme consequence of $(d-1)$-form symmetries which we will use extensively in the context of two dimensional σ-models in paper IV.

Part I:
Low energy features of four dimensional theories

2. Supersymmetry and moduli spaces

2.1 The supersymmetry algebra

It is well known that quantum field theories in d-spacetime dimensions possess a spacetime symmetry group called the ***Poincaré group*** $\mathrm{ISO}(d)$. Its corresponding Lie algebra $\mathfrak{iso}(d)$ is generated by translations P_μ and Lorentz transformations $M_{\mu\nu}$ subject to the commutation relations

$$\begin{aligned}[P_\mu, P_\nu] &= 0,\\ [M_{\mu\nu}, P_\sigma] &= -\mathrm{i}(\eta_{\mu\sigma}P_\nu - \eta_{\mu\sigma}P_\mu),\\ [M_{\mu\nu}, M_{\sigma\xi}] &= -\mathrm{i}(\eta_{\mu\sigma}M_{\nu\xi} - \eta_{\mu\xi}M_{\nu\sigma} - \eta_{\nu\sigma}M_{\mu\xi} + \eta_{\nu\xi}M_{\mu\sigma}),\end{aligned} \tag{2.1}$$

where is $\eta_{\mu\nu}$ the Minkowski metric in the mostly plus convention. If one wishes to introduce supersymmetry, then one has to extend $\mathfrak{iso}(d)$ to a $\mathbb{Z}_2$-graded Lie algebra by introducing anti-commuting generators. The resulting structure is called a ***Poincaré superalgebra***. We will outline this construction in $d = 4$ and refer the interested reader to [22] for details of the higher dimensional construction.

Concretely, a $\mathbb{Z}_2$-graded Lie algebra is a vector space $\mathfrak{l}$ with a decomposition

$$\mathfrak{l} = \mathfrak{l}_0 \oplus \mathfrak{l}_1, \tag{2.2}$$

together with a product $[\cdot,\cdot\} : \mathfrak{l} \times \mathfrak{l} \to \mathfrak{l}$ which satisfies

$$\begin{aligned}&[\mathfrak{l}_i, \mathfrak{l}_j\} \in \mathfrak{l}_{i+j \pmod 2},\\ &[\mathfrak{l}_i, \mathfrak{l}_j\} = (-1)^{ij+1}[\mathfrak{l}_j, \mathfrak{l}_i\},\\ [\mathfrak{l}_i,[\mathfrak{l}_j,\mathfrak{l}_k\}\}(-1)^{ik} + &[\mathfrak{l}_j,[\mathfrak{l}_k,\mathfrak{l}_i\}\}(-1)^{ij} + [\mathfrak{l}_k,[\mathfrak{l}_i,\mathfrak{l}_j\}\}(-1)^{jk} = 0.\end{aligned} \tag{2.3}$$

In order to produce a Poincaré superalgebra, we set introduce a certain number of anti-commuting generators to extend $\mathfrak{iso}(4)$ in a consistent manner. The only consistent way to do this is to introduce generators that lie in the fundamental spinor representation of $\mathrm{O}(3,1)$ [23, 24]. We therefore introduce spinors Q^I_α and $\bar{Q}^I_{\dot\alpha}$ with $I = 1, \dots, \mathcal{N}$ to fulfil this role. These generators must also satisfy

$$\begin{aligned}\{Q^I_\alpha, \bar{Q}^J_{\dot\alpha}\} &= 2\delta^{IJ}\sigma^\mu_{\alpha\dot\alpha}P_\mu,\\ \{Q^I_\alpha, Q^J_\beta\} &= \varepsilon_{\alpha\beta}Z^{IJ},\end{aligned} \tag{2.4}$$

where Z^{IJ} is a set of scalars called the central charges of the algebra and σ^μ is the vector of Pauli matrices. It now remains to describe the commutation relations between the Q^I_α and the generators of the Poincaré algebra. One can deduce that

$$\begin{aligned}
[P_\mu, Q^I_\alpha] &= 0, \\
[M_{\mu\nu}, Q^I_\alpha] &= \mathrm{i}(\sigma_{\mu\nu})_\alpha^\beta Q^I_\beta,
\end{aligned} \tag{2.5}$$

while the commutators involving $\bar{Q}^I_{\dot\alpha}$ can be obtained from Hermitian conjugation. Mathematically, the resulting superalgebra is a group contraction of $\mathfrak{osp}(\mathcal{N}|4)$, a Lie superalgebra with $\mathfrak{l}_0 = \mathfrak{sp}(4) \oplus \mathfrak{o}(\mathcal{N})$.

Remark. Note that there could be an action of the internal symmetry group on the Q^I_α through $Q^I_\alpha \mapsto R^I{}_J Q^J_\alpha$. This is called the *R-symmetry* group and it is isomorphic to $\mathrm{U}(\mathcal{N})$ when the central charges are trivial. When there are non-trivial central charges, this group is reduced to $\mathrm{USp}(\mathcal{N})$.

An immediate consequence of enlarging the symmetry algebra is that fields must now organise into representations of the Poincaré superalgebra. These representations are called *super-multiplets.* Let us specialise to the case of four dimensional $\mathcal{N} = 2$ theories to showcase this construction.

First note that since Z^{IJ} is antisymmetric, there is only one central charge Z given by

$$Z' = \begin{pmatrix} 0 & Z \\ -Z & 0 \end{pmatrix}. \tag{2.6}$$

We can now form ladder operators from the supercharges Q^I_α as

$$\begin{aligned}
a_\alpha &= \frac{1}{\sqrt{2}}\left(Q^1_\alpha + \varepsilon_{\alpha\beta}(Q^2_\beta)^\dagger\right), \\
b_\alpha &= \frac{1}{\sqrt{2}}\left(Q^1_\alpha - \varepsilon_{\alpha\beta}(Q^2_\beta)^\dagger\right).
\end{aligned} \tag{2.7}$$

The super-Poincaré algebra then tells us that the only non-trivial anti-commutators are

$$\begin{aligned}
\{a_\alpha, (a_\beta)^\dagger\} &= (2m + Z)\delta_{\alpha\beta}, \\
\{b_\alpha, (b_\beta)^\dagger\} &= (2m - Z)\delta_{\alpha\beta},
\end{aligned} \tag{2.8}$$

where m is the mass operator. Now to form representations of the $\mathcal{N} = 2$ supersymmetry algebra we need to choose a highest weight state $|\lambda\rangle$, typically called the *Clifford vacuum*, which is annihilated by both a_α and b_α. We then use the creation operators $(a_\alpha)^\dagger$ and $(b_\alpha)^\dagger$ together with their commutation relations to build the full super-multiplet of size $2^4 = 16$.

Note that the by demanding that any state in the Hilbert space has non-negative norm requires that $2m \geq Z$. This is called the ***Bogomol'nyi-Prasad-Sommerfield bound*** or BPS bound for short. When the BPS bound is saturated the super-multiplet truncates and we get a multiplet with fewer than 16 states and the representation theory is equivalent to those of massless states.

A consequence of the Coleman-Mandula theorem [25] is that non-gravitating QFTs cannot possess degrees of freedom with spin greater than 1. From these considerations, it is easy to see that only theories in six or fewer spacetime dimensions satisfy this criteria. If we allow for gravitating theories, and therefore allow *supergravity (SUGRA) theories*, then Coleman-Mandula tells us that spins no higher than two are allowed. Repeating the same analysis shows that SUGRAs exist in eleven and fewer spacetime dimensions.

2.2 The superconformal algebra

A hallmark feature of QFT is the fact that one can deform a theory to trigger a renormalisation group (RG) flow to another theory in the low energy regime [26, 27]. The theory at this fixed point now has an enhanced spacetime symmetry group given by transformations that leaves the metric invariant up to a scaling factor. That is, any transformation with the effect

$$\eta_{\mu\nu}(x) \mapsto g_{\mu\nu}(x') = \Lambda(x)\eta_{\mu\nu}(x). \tag{2.9}$$

All transformations in the Poincaré group trivially satisfy this, but there are now two additional allowed transformations: *dilatations* and *special conformal transformations* [28]. The dilatations act as

$$x^\mu \mapsto e^\lambda x^\mu, \tag{2.10}$$

while the special conformal transformations are given by

$$x^\mu \mapsto \frac{x^\mu - b^\mu(x \cdot x)}{1 - 2(b \cdot x) + (b \cdot b)(x \cdot x)}. \tag{2.11}$$

Now let D and K_μ denote the generators of these transformations respectively. By expanding the above transformations to first order in their transformation parameters we can easily obtain representations for D and K_μ to get the following additional commutation relations

$$\begin{aligned}
[D, P_\mu] &= \mathrm{i}P_\mu, \\
[D, K_\mu] &= -\mathrm{i}K_\mu, \\
[K_\mu, P_\nu] &= 2\mathrm{i}(\eta_{\mu\nu}D - M_{\mu\nu}), \\
[K_\rho, M_{\mu\nu}] &= \mathrm{i}(\eta_{\rho\mu}K_\nu - \eta_{\rho\nu}K_\mu),
\end{aligned} \tag{2.12}$$

d	$\mathfrak{l}$	$\mathfrak{l}_0$	R-symmetry
3	$\mathfrak{osp}(\mathcal{N}\|4)$	$\mathfrak{o}(3,2)\oplus\mathfrak{o}(\mathcal{N})$	$\mathrm{SO}(\mathcal{N})$
4	$\mathfrak{su}(2,2\|\mathcal{N})$ $\mathfrak{psu}(2,2\|4)$	$\mathfrak{o}(4,2)\oplus\mathfrak{su}(\mathcal{N})\oplus\mathbb{C}$ $\mathfrak{o}(4,2)\oplus\mathfrak{su}(\mathcal{N})$	$\mathrm{U}(\mathcal{N})$ if $\mathcal{N}\neq 4$ $\mathrm{SU}(\mathcal{N})$ if $\mathcal{N}=4$
5	$\mathfrak{f}(4)$	$\mathfrak{o}(5,2)\oplus\mathfrak{su}(2)$	$\mathrm{SO}(3)$
6	$\mathfrak{osp}(6,2\|2\mathcal{N})$	$\mathfrak{sp}(2\mathcal{N})\oplus\mathfrak{o}(6,2)$	$\mathrm{Sp}(2\mathcal{N})$

Table 2.1. *The possible superconformal algebras in various dimensions.*

while all other commutators involving D or K_μ vanish.

We can simplify the situation by combining all the generators in one large matrix $(d+2)\times(d+2)$ matrix J_{AB}. Concretely, we let

$$J_{\mu\nu}=M_{\mu\nu},\ J_{-1,\mu}=\frac{1}{2}(P_\mu-K_\mu),\ J_{-1,0}=D,\ J_{0,\mu}=\frac{1}{2}(P_\mu+K_\mu), \tag{2.13}$$

and impose antisymmetry $J_{AB}=-J_{BA}$. It is now easy to see that J_{AB} generates $\mathfrak{o}(d,2)$ as a Lie algebra. As such, the conformal algebra in d-spacetime dimensions is, in fact, a simple Lie algebra.

Having established the conformal algebra $\mathfrak{o}(d,2)$, it is now interesting to ask if one can supersymmeterise this to obtain the *superconformal algebras*. The fact that the conformal algebras are actually simple Lie algebras puts a heavy restriction on any resulting supersymmetric extension– they must be simple Lie superalgebras as classified in [29]. To obtain the only possible superconformal algebras we now simply need to impose the condition that the generators of $\mathfrak{l}_1$ transform as spinors of the conformal group. This leads to table 2.1 which details the possible superconformal algebras in dimensions $3\leq d\leq 6$. Note that superconformal algebras only exist in dimensions 6 and lower [24,30,31], a direct consequence of the classification of simple Lie superalgebras.

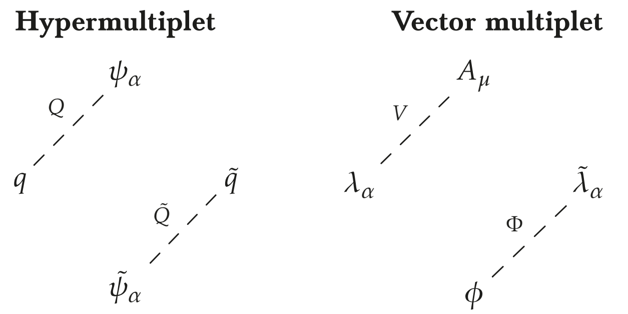
One can also analyse the possible matter content of conformal theories through its representations. This was done in the four dimensional context in [32] and more generally in [18].

Remark. Note that the constructions above allows for both supersymmetry and superconformal algebras with $\mathcal{N}=3$ in four dimensions. Despite this, it was long thought that any theory that has $\mathcal{N}=3$ supersymmetry in four dimensions was actually an $\mathcal{N}=4$ theory. This is due to the fact that the only free multiplet in $\mathcal{N}=3$ is identical to that of a vector multiplet in $\mathcal{N}=4$, therefore immediately ruling out any weakly coupled theory with only $\mathcal{N}=3$ supersymmetry. Recently, however, strongly coupled examples of $\mathcal{N}=3$ theories have been constructed geometrically [33].

2.3 Branches of the moduli space

Having established the basics of supersymmetric field theories, we now wish to build some theories and examine their dynamics. In particular, we will focus on the vacuum configurations of four dimensional $\mathcal{N} = 2$ theories. For a more detailed account of this material, we refer the interested readers to [34, 35].

In these types of theories, there are only two types of super-multiplets that we can use: the vector multiplet and the hypermultiplet. We can present them in the following manner.



Here the $\mathrm{SU}(2)_R \subset \mathrm{U}(2)_R$ symmetry acts on the rows of the multiplets and we've partitioned the fields into $\mathcal{N} = 1$ super-multiplets. Concretely, we have that the hypermultiplet can be seen as two $\mathcal{N} = 1$ chiral multiplets $(Q, \tilde{Q})$ and the vector multiplet can be seen as a chiral multiplet Φ with an $\mathcal{N} = 1$ vector multiplet V.

Now let us consider the most general $\mathcal{N} = 2$ Lagrangian we can make with a vector multiplet and some, possibly massive, matter hypermultiplets. Given a vector multiplet (Φ, V), there are two types of terms that can appear in the Lagrangian. These are given by

$$\mathcal{L}_{\text{gauge}} = \frac{1}{4\pi}\mathrm{Im}\left[\int \mathrm{d}^2\theta\, \mathrm{d}^2\bar{\theta}\, \mathrm{tr}\left(\Phi^\dagger e^V \Phi\right) + \frac{\tau}{2}\int \mathrm{d}^2\theta\, \mathrm{tr}\, W_\alpha W^\alpha\right]. \tag{2.14}$$

The first term, integrated over all of superspace, is a straightforward D-term, but the second term requires a little more explanation. First of all, the factor τ appearing is the *holomorphic gauge coupling*

$$\tau = \frac{\vartheta}{2\pi} + \frac{4\pi \mathrm{i}}{g^2}, \tag{2.15}$$

which encodes both the ϑ-angle and the Yang-Mills gauge coupling g. Secondly, the field W_α is not independent of V. Indeed, we have the expression

$$W_\alpha = -\frac{1}{4}\overline{DD}D_\alpha(V), \tag{2.16}$$

where D and $\bar{D}$ are the usual covariant derivatives on superspace. It is easy to check that it is a chiral superfield, so the combination $\operatorname{tr} W_\alpha W^\alpha$ can be integrated over only half of superspace.

If we introduce M many hypermultiplets $(Q_i, \tilde{Q}^i)$, we now have several additional terms we can add to the Lagrangian. The first is the mixed F-terms between the hypermultiplet fields and the vector multiplet fields

$$\mathcal{L}_{\text{mixed}} = \frac{1}{4\pi}\operatorname{Im}\left[\int \mathrm{d}^2\theta\, \mathrm{d}^2\bar{\theta}\, \operatorname{tr}\left(Q_i^\dagger e^V Q^i + \tilde{Q}^{\dagger i} e^V \tilde{Q}_i\right)\right]. \tag{2.17}$$

Note that here we are summing over repeated indices to include several hypermultiplets. Finally, we can add an $\mathcal{N} = 1$ superpotential to give mass to the hypermultiplets. In fact, $\mathcal{N} = 2$ supersymmetry fixes this term to be

$$\mathcal{L}_{\text{pot.}} = \frac{1}{4\pi}\operatorname{Im}\left[\tau\sqrt{2}\int \mathrm{d}^2\theta\, \left(\tilde{Q}_i \Phi Q^i + m^i_j \tilde{Q}_i Q^j\right)\right]. \tag{2.18}$$

Demanding $\mathrm{SU}(2)_R$ symmetry on this term requires the matrix $m = (m^j_i)$ to satisfy $[m, m^\dagger] = 0$. As such, it can be diagonalised and we could change basis to simplify this term. For now, we will keep the Lagrangian in this more general form. Overall, we have that the most general $\mathcal{N} = 2$ preserving Lagrangian is given by

$$\begin{aligned}
\mathcal{L} &= \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{mixed}} + \mathcal{L}_{\text{pot.}} \\
&= \frac{1}{4\pi}\operatorname{Im}\left[\int \mathrm{d}^2\theta\, \mathrm{d}^2\bar{\theta}\, \operatorname{tr}\left(\Phi^\dagger e^V \Phi + Q_i^\dagger e^V Q^i + \tilde{Q}^{\dagger i} e^V \tilde{Q}_i\right)\right] \\
&+ \frac{1}{4\pi}\operatorname{Im}\left[\tau \int \mathrm{d}^2\theta\, \frac{1}{2}\operatorname{tr} W_\alpha W^\alpha + \sqrt{2}\left(\tilde{Q}_i \Phi Q^i + m^i_j \tilde{Q}_i Q^j\right)\right]
\end{aligned} \tag{2.19}$$

It is worth noting that the first term is typically called the ***Kähler potential*** K. We will revisit its importance shortly.

Equipped with the general Lagrangian eq. (2.19), we can now look at possible vacuum configurations of the system. The D-term equations of motion are given by

$$\begin{aligned}
[\phi, \phi^\dagger] &= 0, \\
[q^i q_i^\dagger - \tilde{q}_i \tilde{q}^{\dagger i}]_{\text{traceless}} &= 0.
\end{aligned} \tag{2.20}$$

While the F-term equations are

$$\begin{aligned}
[q^i \tilde{q}_i]_{\text{traceless}} &= 0, \\
q^i m^j_i + \phi q^j &= 0, \\
m^i_j \tilde{q}_i + \tilde{q}^j \phi &= 0.
\end{aligned} \tag{2.21}$$

The solution space to these equations forms the *moduli space* of the theory, but there are particular types of solution we should note. Here we take the hypermultiplet masses m_i^j to vanish for simplicity.

Coulomb branch. A simple set of solutions is given by taking $q^i = \tilde{q}_i = 0$ and ϕ non-zero. The D-term equation of motion then tells us that ϕ is a normal matrix and can therefore be diagonalised. This means that the effective theory on the Coulomb branch is a $\mathrm{U}(1)^r$ gauge theory, where r is the rank of the gauge group. For example, if $G = \mathrm{SU}(4)$ then the effective theory is a $\mathrm{U}(1)^3$ gauge theory.

Higgs branch. Instead taking $\phi = 0$ and the hypermultiplet scalars non-zero gives us the so-called Higgs branch. The effective theory on this branch is simply the theory of free hypermultiplets since the gauge group is typically fully Higgsed here (hence the name).

One reason for singling out these two branches of the moduli space is the fact that supersymmetry heavily restricts their geometry. The following theorem is an important example of this.

Theorem 2.1. (Freed [36]). The Coulomb branch of a four dimensional $\mathcal{N} = 2$ theory is a special Kähler manifold.

Recall that a Kähler manifold is a symplectic manifold (M, ω) equipped with an almost complex structure J such that

$$g(u, v) = \omega(u, Jv), \tag{2.22}$$

is a Riemannian metric on M. A *special Kähler structure* on M is a real flat torsion free connection ∇ that satisfies

$$\mathrm{d}_\nabla J = 0. \tag{2.23}$$

The existence of such a connection gives rise to a set of *special coordinates* on M. In particular, there exist coordinates a^i along with dual variables a_i^D such that the Kähler potential K is given by

$$K = \frac{1}{2}\mathrm{Im}\left(a_i^D \bar{a}^i\right) \tag{2.24}$$

and the dual variables are related through

$$a_i^D = \frac{\partial \mathcal{F}}{\partial a^i}, \tag{2.25}$$

where $\mathcal{F}$ is a function called the *prepotential*. The physical significance of this theorem is that the low energy theory on the Coulomb branch is completely determined by the prepotential. Indeed, not only are the dual variables related through $\mathcal{F}$, we also have that the holomorphic gauge coupling

τ can be written as

$$\tau_{ij} = \frac{\partial^2 \mathcal{F}}{\partial a^i \partial a^j} = \frac{\partial a_i^D}{\partial a^j}. \tag{2.26}$$

By specialising eq. (2.19) to a $\mathrm{U}(1)^r$ gauge theory, we can easily rewrite the resulting action in terms of the prepotential. Doing so gives us

$$\mathcal{L}_{\text{eff.}} = \frac{1}{4\pi}\mathrm{Im}\left[\int \mathrm{d}^4\theta \, \frac{\partial \mathcal{F}}{\partial \Phi^i}\bar{\Phi}^i + \frac{1}{2}\int \mathrm{d}^2\theta \, \frac{\partial^2 \mathcal{F}}{\partial \Phi^i \partial \Phi^j} W_\alpha^i W^{\alpha j}\right], \tag{2.27}$$

where the Φ^i denotes the chiral supermultiplet part of each $\mathrm{U}(1)$ vector multiplet.

Finally, it is worth noting that the central charge function also simplifies when written in terms of the special coordinates. For a state with electric charges q_i, magnetic charges n_i and flavour charges f_i, the central charge is given by

$$Z(\mathbf{q}, \mathbf{n}) = \sum_i (q_i a_i + n_i a_i^D) + \sum_j M_j f_j \tag{2.28}$$

where M_j are the eigenvalues of the mass matrix m_i^j appearing in the UV Lagrangian.

While we won't consider the Higgs branch much in this thesis, it is worth noting that the structure of the Higgs branch is equally as rich. In fact, the Higgs branch is always an example of a *hyper-Kähler* manifold [37, 38].

2.4 Wall-crossing

Topologically, the Coulomb branch of a theory is simply $\mathbb{C}^r$. However, this doesn't tell the full story, since we've seen that the Coulomb branch is metrically much more interesting. One important point is that the Coulomb branch could have metric singularities. These singular points have a simple physical meaning which allows us to infer information about the theory's BPS spectrum.

Consider an $\mathcal{N} = 2$ theory defined at some large energy scale Λ. If we begin to lower this scale and integrate out appropriately massive states, then we will eventually integrate out all massive states and only be left with vacuum states. In other words, we have moved onto the moduli space of the theory. The singular points of the Coulomb branch then arise from the fact that we may have integrated out states that become massless at certain values of the moduli [3]. As such, these states are BPS states! However, these singularities don't give us the full BPS spectrum. Indeed, it is possible for these states to form stable bound states which would not correspond to a singularity of the Coulomb branch. An interesting question to ask now is, if we know the singular points of the Coulomb branch, what is the full BPS spectrum? This turns out to be a subtle question, which we will now illustrate in a simple example.

Example. Consider pure $\mathcal{N} = 2$ SU(2) gauge theory. Moving to the Coulomb branch allows us to to diagonalise the bottom component of the vector multiplet ϕ as

$$\phi = \begin{pmatrix} a & 0 \\ 0 & -a \end{pmatrix}. \tag{2.29}$$

We can therefore give a gauge-invariant coordinate on the Coulomb branch as $u = a^2$, up to quantum corrections. In order to determine the variable dual to a, we note that the one-loop running of τ is known to be [34]

$$\tau(a) = -\frac{8}{2\pi \mathrm{i}} \log \frac{a}{\Lambda} + \dots \tag{2.30}$$

where Λ is the defining energy scale. By using eq. (2.26) we can obtain the a^D by integrating $\tau(a)$ to get

$$a^D = -\frac{8a}{2\pi \mathrm{i}} \log \frac{a}{\Lambda} + \dots \tag{2.31}$$

If we stay in the weak coupling regime of $|a| \gg |\Lambda|$, we can use the leading order terms as a good approximation. Now consider moving along the path $u = re^{\mathrm{i}\phi}$ with $\phi \in [0, 2\pi)$ with r large. Since $u = a^2$ to first order, have that

$$(a, a^D) \mapsto (a, a^D) \begin{pmatrix} -1 & 4 \\ 0 & -1 \end{pmatrix}. \tag{2.32}$$

This tells us that there is a non-trivial monodromy at infinity. The presence of this alone would signal that there would be some form of logarithmic singularity in the Coulomb branch, which wouldn't make much sense physically. Therefore, we must have some singular points on the Coulomb branch to source local monodromies with the same overall effect on the variables. It was shown [3] that in the strong coupling regime, there are two singular points which do exactly this. Furthermore, it was shown that the corresponding BPS states are given by a dyon of charge $(2, -1)$ and a monopole of charge $(0, 1)$. However, this is not the full spectrum everywhere in the Coulomb branch. At weak coupling there can be an effective theta angle, which allows a Witten effect to occur [39]. This generates a tower of dyons with charges $(2n, 1)$ and $(2n + 2, -1)$ together with a W boson of charge $(2, 0)$. At strong coupling, we cannot have an effective theta angle, thus leaving us with only the monopole and the dyon from the singularities.

The example above illustrates an important point: the stable BPS spectrum is not constant over the Coulomb branch. In fact, the best we can say is that it is *locally* constant and the spectrum jumps when certain submanifolds are crossed. This phenomenon is called *wall-crossing*.

To understand why BPS states undergo wall-crossing, we note that by the linearity of the central charge function we have

$$|Z(\gamma+\gamma')| \leq |Z(\gamma)| + |Z(\gamma')|, \tag{2.33}$$

where γ and γ' are the charges of two putative BPS states. From this equation we see that if the bound state with charge $\gamma+\gamma'$ is stable, then, by the BPS bound, it is energetically favoured over γ and γ' separately. By varying parameters in the moduli space we are able to find codimension-1 submanifolds where this inequality is saturated. This occurs whenever we have

$$\arg Z(\gamma) = \arg Z(\gamma'). \tag{2.34}$$

At such points, bound states of γ and γ' can either form or decay. The submanifolds $\mathcal{W}(\gamma,\gamma')$ defined by the above equality are called *walls of marginal stability* or *walls of the first kind*. The walls of marginal stability partition the Coulomb branch into chambers where the BPS spectrum differs through wall-crossing transitions. Each chamber is typically referred to as a *BPS chamber*.

With the walls of marginal stability identified, we now wish to know exactly how the spectrum changes under a wall-crossing transition. While the full behaviour of wall-crossing transitions was found by Kontsevich and Soibelman in [40,41], there were several primitive results on the matter before their result [42–44]. For the sake of brevity, we will only discuss the full solution.

Let Γ be the lattice of BPS charges of a theory. To each $\gamma \in \Gamma$ we can associate a formal variable X_γ which generates an auxiliary algebra called the *quantum torus algebra* $\mathbb{T}_\Gamma(q)$. We can characterise $\mathbb{T}_\Gamma(q)$ by stating the algebra product to be

$$X_\gamma X_{\gamma'} = q^{\langle\gamma,\gamma'\rangle/2} X_{\gamma+\gamma'} = q^{\langle\gamma,\gamma'\rangle} X_{\gamma'} X_\gamma \tag{2.35}$$

where $\langle\cdot,\cdot\rangle$ is a symplectic form which we take to be the Dirac pairing of states and q is a formal variable. We also define the *q-exponential function* as

$$E_q(z) = \prod_{m=0}^{\infty} \frac{1}{1+q^{m+1/2}z} = 1 + \sum_{n=1}^{\infty} \frac{(-1)^n q^{n/2} z^n}{\prod_{k=1}^{n}(1-q^k)}. \tag{2.36}$$

From this we can associate to each $\gamma \in \Gamma$ a formal power series in q given by

$$U_\gamma = \prod_{n\in\mathbb{Z}} E_q\left((-1)^n q^{n/2} X_\gamma\right)^{(-1)^n \Omega_n(\gamma)}, \tag{2.37}$$

where $\Omega_n(\gamma)$ are the coefficients of the protected-spin character of γ charged states. The exact form of $\Omega_n(\gamma)$ will not matter much for our discussion, so we refer the interested reader to the original paper [45] for a in depth discussion. We will simply note that if we have a hypermultiplet of charge γ then

$$\Omega_n(\gamma) = \begin{cases} 1, & \text{if } n = 0, \\ 0, & \text{otherwise.} \end{cases} \tag{2.38}$$

Finally, let $\mathcal{B}(p)$ denote the set of stable BPS charges at a point p in the moduli space. Then the operator

$$\mathcal{O}(q) = \prod_{\gamma\in\mathcal{B}(p)}^{\curvearrowleft} U_\gamma, \tag{2.39}$$

where the product is taken in order of increasing $\arg Z(\gamma)$, is invariant under wall-crossing [40, 41]. In particular, calculation of $\mathcal{O}(q)$ from any BPS chamber yields the same invariant.

The fact that eq. (2.39) is invariant under wall-crossing gives us a lot of information. Let us make a few remarks about this.

1. First of all, if X_γ and $X_{\gamma'}$ commute we observe no additional terms appearing in $\mathcal{O}(q)$ upon commuting the q-exponentials. This tells us that states with $\langle\gamma,\gamma'\rangle = 0$ do not undergo non-trivial wall-crossing transitions with each other. As such, the Dirac pairing between two states dictates the type of wall-crossing transition that will occur.
2. In the case of $\langle\gamma,\gamma'\rangle = 1$, it is easy to check that

$$E_q(X_\gamma)E_q(X_{\gamma'}) = E_q(X_{\gamma'})E_q(X_{\gamma+\gamma'})E_q(X_\gamma). \tag{2.40}$$

   This is expected, as this is the known wall-crossing transition for the A_2 Argyres-Douglas theory [46].
3. Similarly, one can use the definition of $\mathcal{O}(q)$ to check the wall-crossing transition of pure SU(2) gauge theory as described above.
4. The operator $\mathcal{O}(q)$ also gives us other information about the theory. For example, in [47] it was conjectured that one can obtain the Schur index of a theory by taking a generalised trace of $\mathcal{O}(q)$ and weighting it appropriately.

In theory, the Kontsevich-Soibelman operator encodes all possible types of wall-crossing, but we should note that there is a caveat. If $\langle\gamma,\gamma'\rangle \geq 3$ then the wall-crossing identity is not known. Chambers where such states appear are called *wild* and occur often [48].

Remark. While we have discussed wall-crossing in a purely physical sense, it should be noted that wall-crossing is familiar to mathematicians in a different, but related, context. It turns out the Donaldson-Thomas invariants of certain manifolds display the same wall-crossing behaviour [49].

2.5 BPS quivers

We have seen that understanding wall-crossing comes down to understanding how to manipulate the Kontsevich-Soibelman operator $\mathcal{O}(q)$. While this is conceptually simple, in practice this can be difficult. Furthermore, in order to write it down we need knowledge of at least one BPS chamber. It is therefore natural to ask if we can find a tool for finding the full BPS chambers of a theory without resorting to manipulating $\mathcal{O}(q)$.

For now, suppose we know about at least one chamber of a theory. We can therefore form the charge lattice of the theory Γ and, by CPT symmetry, we can decompose it as

$$\Gamma = \Gamma^+ \cup \Gamma^-, \quad \Gamma^- = -\Gamma^+ \tag{2.41}$$

by imposing that the $Z(\gamma)$ for $\gamma \in \Gamma^+$ lie in a particular half-plane. Physically, this is splitting the spectrum into states and anti-states. There are many ways of doing this, but for now any such decomposition will suffice.

Now we suppose can find a positive integral basis for Γ^+. If we denote this basis as $\{\gamma_i\}_{i=1}^N$, then we can form the matrix

$$B_{ij} = \langle \gamma_i, \gamma_j \rangle. \tag{2.42}$$

We can represent this graphically by drawing a node for element γ_i of the basis and then connecting node i to node j by $\max(B_{ij}, 0)$ many arrows. This forms a directed graph which we will call the ***BPS quiver*** of the theory and B the associated ***intersection matrix***.

Example. Consider pure SU(2) gauge theory. We saw in the previous section that there are two states with charges $\gamma_1 = (2, -1)$ and $\gamma_2 = (0, 1)$ that provide a basis for all BPS charges. The corresponding BPS quiver is given by

$$\gamma_1 \rightrightarrows \gamma_2$$

Presented in this fashion, the BPS quiver is nothing more than a summary of the possible BPS contents of the theory and their Dirac pairings. However, it was shown in [5] that the BPS quiver encodes much more information. We will review their construction.

Given a quiver we now wish to ask if there exists a BPS state of charge γ in the spectrum. Since we have a positive integral basis of the charge lattice we decompose γ as

$$\gamma = \sum_i n_i \gamma_i, \tag{2.43}$$

where the $n_i \in \mathbb{Z}_{\geq 0}$. We now introduce an ancillary quiver quantum mechanics based on the BPS quiver. Concretely, we replace each node i with a gauge group $\mathrm{U}(n_i)$ and introduce bifundamental matter B^a_{ij} charged under $\mathrm{U}(n_i)$ and $\mathrm{U}(n_j)$ for each arrow a connecting $i \to j$. We now supplement the quiver with a superpotential $W(B^a_{ij})$ whose terms correspond to loops in the quiver. The main observation of [5] is that asking whether there exists a BPS state of charge γ in our four dimensional theory is equivalent to seeing if the the moduli space of charge γ ground states $\mathcal{M}_\gamma$ of the quiver quantum mechanics is non-empty[1].

We now wish to characterise $\mathcal{M}_\gamma$ in order to infer the BPS spectrum of the four dimensional theory. There are two ways of doing this: by solving both the D-term and F-term equations of motion or by performing a GIT quotient [50] and trading the D-term equation of motion with a stability condition together with complexifying the gauge group. The latter takes on interesting mathematical meaning, so we will pursue that route.

By complexifying the gauge groups we exchange each $\mathrm{U}(n_i)$ for a $\mathrm{GL}(n_i,\mathbb{C})$ factor instead. The bifundamentals B^a_{ij} now take on the role of linear maps between the vector spaces $\mathbb{C}^{n_i}$ and $\mathbb{C}^{n_j}$ such that the F-term equations of motion $\partial W/\partial B^a_{ij} = 0$ are satisfied. This is exactly the datum for a ***quiver representation*** for a quiver with a potential [51, 52]. When phrased in this language, the notion of stability becomes quite simple. Let R be a quiver representation defined on vectors spaces $\mathbb{C}^{n_i}$ and maps B^a_{ij}. A representation S defined on spaces $\mathbb{C}^{m_i}$ and maps b^a_{ij} is said to be a ***subrepresentation*** if $m_i < n_i$ for each i and the diagram

$$\begin{tikzcd} \mathbb{C}^{n_i} \arrow[r, "B^a_{ij}"] & \mathbb{C}^{n_j} \\ \mathbb{C}^{m_i} \arrow[u] \arrow[r, "b^a_{ij}"] & \mathbb{C}^{m_j} \arrow[u] \end{tikzcd}$$

commutes for all choices of i, j and a. A ***stable representation*** is a quiver representation R such that, for all proper subrepresentations S, we have

$$\arg\left(\sum_i m_i Z(\gamma_i)\right) < \arg\left(\sum_i n_i Z(\gamma_i)\right). \tag{2.44}$$

This particular stability condition is called Π-stability and was first studied in the context of BPS branes in string theory [53]. More generally, it is an example of ***Bridgeland stability*** on the triangulated category of derived quiver representations [54].

[1]This can be motivated easily from string theory, where the quiver quantum mechanics arises as the dimensional reduction of the D-brane world-volume theory to one dimension. We will discuss the geometric aspects of BPS quivers more in part II.

Finally, the moduli space $\mathcal{M}_\gamma$ with $\gamma = \sum_i n_i \gamma_i$ can be succinctly written as

$$\mathcal{M}_\gamma = \left\{ R = \{B^a_{ij} : \mathbb{C}^{n_i} \to \mathbb{C}^{n_j}\} \;\middle|\; \frac{\partial W}{\partial B^a_{ij}} = 0,\ R \text{ is } \Pi-\text{stable} \right\} \Big/ \prod_i \mathrm{GL}(n_i, \mathbb{C}), \tag{2.45}$$

where $R = \{B^a_{ij} : \mathbb{C}^{n_i} \to \mathbb{C}^{n_j}\}$ is a representation of the BPS quiver. Therefore, the existence of a BPS state with charge γ is equivalent to finding a stable quiver representation of the BPS quiver at that point of the moduli space. Furthermore, as explained in [5], one can infer the spin and R-charge through considerations of this space.

One of the most remarkable things about the quiver representation point of view is that it gives rise to a simple algorithm for determining BPS chambers. The algorithm, first presented in [5], relies on a quiver operation called *mutation*. To describe this we first have to add some additional data to our quiver.

To each node i we can associate a *cluster variable* $c^0_i = \gamma_i$ which we will keep track of at every step of the algorithm by increasing the upper index. Mutating the quiver at a node k then enforces the shift

$$c^{t+1}_i = \mu_k(c^t_i) = \begin{cases} c^t_i + c^t_k \max[0, \langle c^t_i, c^t_k \rangle], & \text{if } i \neq k, \\ -c^t_i, & \text{otherwise.} \end{cases} \tag{2.46}$$

The new form of the quiver can then be found by taking the Dirac pairings of the transformed cluster variables. It is easy to see that this has the effect of reversing the direction of any arrow point to or from node k any completing and 3-cycles $a \to k \to b$ after the reversal of arrows.

Physically, mutation has an interpretation as a new kind of wall-crossing. Recall that we chose an arbitrary decomposition $\Gamma = \Gamma^+ \cup \Gamma^-$ by specifying a half-plane where the central charges of $\gamma \in \Gamma^+$ lie. We could imagine tilting this half plane as in fig. 2.1 so the state with largest argument leaves the half-plane defining Γ^+. This then gives rise to a new positive integral basis, as specified by the mutated basis given above. This type of transition can occur if we move in the moduli space and cross a new type of wall– a *wall of the second kind.*

With this tilting picture, we can see that we can probe the whole spectrum if we continually tilt the half-plane until we arrive at the CPT conjugates. The mutation algorithm can now be stated as follows.

1. Choose an ordering for the arguments of $Z(\gamma_i)$. Here we are using the convention that we measure arguments in the range $(-\pi, \pi]$.
2. Mutate the node corresponding to the charge with the largest argument and update all cluster variables.

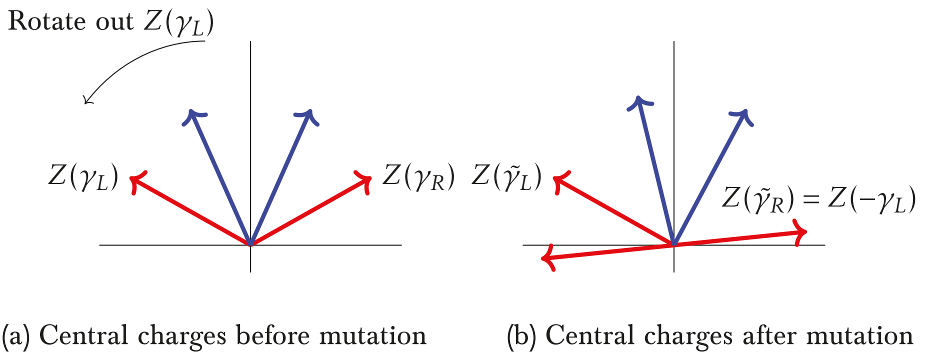


Figure 2.1. Rotation of a central charge out of the 'particle' half-plane requires a relabelling of the BPS quiver. This is enforced by mutation on the node whose charge passes out of the half-plane.

3. Continue to mutate on the nodes whose cluster variable has the largest argument and record the sequence $\mathbf{k}$ of mutated notes. Such a sequence is called a *maximally green sequence* in the literature [55].
4. Once all cluster variables are equal to negative the original charges, the algorithm is complete.

Then the BPS chamber corresponding the ordering chosen in step 1 is given by hypermultiplets with charges $\{c^t_{k(t)}\}$ together with the CPT conjugates.

The mutation algorithm has been successfully applied to find BPS chambers for a wide variety of theories [5, 19, 47, 56–60]. However, if a BPS chamber contains states that aren't hypermultiplets then there are issues. As seen in the weak coupling case of SU(2) calculated in [5], vector multiplets arise as an *accumulation ray* of towers of hypermultiplets. If there is only one such ray then the spectrum can be deduced, but if there are more then there is a region of the spectrum that cannot be probed by mutation. This is the case for wild quivers.

Remark. Throughout this section we have assumed that we at least had some knowledge about a BPS chamber of our theory in order to construct the BPS quiver. However, it is often the case that one can derive the BPS quiver purely geometrically from string theory. Indeed, one of the key points of papers I and II are to obtain these quivers by purely geometric means.

2.6 Maruyoshi-Song flows

Finally, let us end this chapter by discussing an interesting phenomenon that can occur in $\mathcal{N} = 2$ theories.

Given a theory with Lagrangian $\mathcal{L}$, one can trigger an RG flow by adding a relevant deformation $\delta\mathcal{L}$ to the Lagrangian and giving some of the fields which $\delta\mathcal{L}$ depends on a non-zero expectation value. Of course, adding a deformation could in principle break some of the symmetry of the system. For example, if $\delta\mathcal{L}$ is made up of $\mathcal{N}=1$ superfields, then we have explicitly broken the supersymmetry to $\mathcal{N}=1$ and the IR theory is only guaranteed to have $\mathcal{N}=1$ supersymmetry. However, for certain types of $\mathcal{N}=1$ deformations, the supersymmetry in the IR can enhance to $\mathcal{N}=2$ again. Such supersymmetry enhancing flows were first presented in [61, 62] and are often called *Maruyoshi-Song flows.*

Let us review the construction of [61, 62]. Suppose we have a theory with a Lagrangian $\mathcal{L}$ and continuous flavour symmetry group F. By definition, there is an associated conserved flavour current multiplet $\mathcal{F}$ whose bottom component is a scalar μ called the *moment map operator.* We now deform the theory through

$$\mathcal{L} \mapsto \mathcal{L} + \alpha \operatorname{tr}(\mu M), \tag{2.47}$$

where M is an $\mathcal{N}=1$ chiral multiplet. This clearly breaks the supersymmetry to $\mathcal{N}=1$, but if we give M a nilpotent vacuum expectation value, it may happen to restore $\mathcal{N}=2$ supersymmetry in the IR. It is known, via the Jacobson-Morozov theorem [63, 64], that if $\mathfrak{f} = \mathrm{Lie}(F)$ is a semi-simple Lie algebra then any nilpotent element of $\mathfrak{f}$ can be obtained from an embedding $\rho : \mathfrak{su}(2) \hookrightarrow \mathfrak{f}$. Furthermore, the nilpotent element in $\mathfrak{f}$ is the image of

$$\sigma^+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \tag{2.48}$$

under ρ. Giving M such a nilpotent expectation value explicitly breaks part of the flavour symmetry. Indeed, the unbroken part of the flavour algebra is given by the commutatant of $\rho(\sigma^+)$ inside $\mathfrak{f}$.

Note that the nilpotent orbits of semi-simple Lie algebras are well studied [65]. In the case of $F = \mathrm{SU}(N)$, it is known that nilpotent orbits are labelled by partitions of N or, equivalently, Young tableaux. Presenting the partition as

$$P(N) = [1^{n_1}, 2^{n_2}, \ldots, N^{n_N}], \tag{2.49}$$

with $\sum_k k n_k = N$, we get that the resulting flavour algebra is given by

$$\mathfrak{f}_{\mathrm{IR}} = \mathfrak{s}\left[\prod_i \mathfrak{u}(n_i)\right], \tag{2.50}$$

where the $\mathfrak{s}[\cdot]$ imposes the overall traceless condition. The nilpotent orbit corresponding to the partition $[N]$ is called the *principal embedding* and completely breaks the flavour symmetry. The case of $F = \mathrm{SO}(2N)$ is similar, as we can label nilpotent orbits by partitions of $2N$ with a condition on the even terms appearing [65]. More generally, one can specify nilpotent orbits through *Bala-Carter labels* [66]. These labels specify particular Levi subalgebras of $\mathfrak{f}$ which contain a representative of the nilpotent orbit and are minimal. We refer readers to [67] for a more in depth discussion.

After triggering the RG flow specified by a nilpotent orbit, there is no guarantee that the theory in the IR does indeed retain $\mathcal{N}=2$ supersymmetry. We therefore need to check several quantities to see if the enhancement is possible. One simple method is to use a-maximisation to calculate the central charges (a,c) of the IR theory and see if they match any known $\mathcal{N}=2$ SCFT. Furthermore, it has been noted that if the IR central charges are irrational, then there is no supersymmetry enhancement [68]. These checks, together with the knowledge of the flavour symmetry, are often enough to identify the IR theory.

Example. Consider $\mathrm{SU}(N)$ gauge theory with $2N$ fundamental hypermultiplets. This is a Lagrangian SCFT with flavour group $\mathrm{U}(2N)$ when $N>2$ and $\mathrm{SO}(8)$ flavour symmetry for $N=2$. The central charges of these theories are well known to be

$$a=\frac{7N^2-5}{24},\quad c=\frac{2N^2-1}{6}. \tag{2.51}$$

Take $N\geq 3$ and write the flavour symmetry as $\mathrm{SU}(2N)\times\mathrm{U}(1)$. Then we can add a chiral multiplet and give it a nilpotent expectation value specified by the principal embedding of $\mathfrak{su}(2)$ into $\mathfrak{su}(2N)$. The IR central charges can be calculated by a-maximisation as

$$a_{\mathrm{IR}}=\frac{12N^2-5N-5}{24(N+1)},\quad c_{\mathrm{IR}}=\frac{3N^2-N-1}{6(N+1)}. \tag{2.52}$$

These exactly match the central charges for the A_{2N-1} Argyres-Douglas theory. We will discuss such theories in more depth next chapter, but it is worth mentioning that this is an $\mathcal{N}=2$ *non-Lagrangian* SCFT. While it possesses no Lagrangian, it is sometimes said that this procedure gives the theory an $\mathcal{N}=1$ Lagrangian.

3. Seiberg-Witten theory

In section 2.3 we saw that supersymmetry imposes heavy restrictions on the moduli space of a theory. On the Coulomb branch this is manifest in the existence of special coordinates and the prepotential $\mathcal{F}$. In [3, 4] this was further developed in to *Seiberg-Witten theory* which allows us to understand the Coulomb branch from the point of view of Riemann surfaces or, more generally, abelian varieties.

3.1 Motivation

Let us briefly motivate why we might expect a link between the Coulomb branch physics and Riemann surfaces.

Recall that one of the hallmark features of the Coulomb branch was the presence of singluarities which signalled the presence of a BPS state in the spectrum. These singularities also sourced monodromies required to ensure that monodromy at infinity is cancelled. A similar effect occurs when considering the homology of complex manifolds. The resulting theory, called *Picard-Lefschetz theory*, is a complex extension of the well known study of Morse functions.

Let S be a Riemann surface of genus g specified by the holomorphic function $f(w,z) = 0$. Further assume that f depends on some moduli u that can deform the surface. As S has genus g, it is easy to see that

$$\mathrm{H}_1(S, \mathbb{Z}) = \mathbb{Z}^{2g}, \tag{3.1}$$

for generic values of u. However, there may be non-generic values for which the homology jumps and the curve becomes singular. This signals the presence of a *vanishing cycle*. For example, this can happen in the case of

$$f(w,z) = w^2 - (z-a_1)(z-a_2)(z-a_3). \tag{3.2}$$

For generic values of the a_i, the surface $S = \{(w,z) \,|\, f(w,z) = 0\}$ is a genus-1 Riemann surface. However, if $a_1 = a_2$, then the torus becomes 'pinched' and the homology reduces to $\mathrm{H}_1(S^*, \mathbb{Z}) = \mathbb{Z}$. This is depicted in fig. 3.1. The presence of this vanishing cycle is not dissimilar to the singular points that appear on the Coulomb branch. It is therefore natural to ask if there is some notion of monodromy that occurs here too.

Figure 3.1. Left: A genus-1 Riemann surface (torus) for generic values of the moduli. *Right:* A 'pinched' Riemann surface that occurs at non-generic values of the moduli. This has the effect of reducing the first homology group and sourcing monodromy.

Returning to the general genus case, let $\{\alpha_i, \beta^i \,|\, i = 1, \dots, g\}$ be the generators of $\mathrm{H}_1(S, \mathbb{Z})$ at generic values of u with intersection numbers

$$\alpha_i \circ \alpha_j = 0, \quad \alpha_i \circ \beta^j = \delta_i^j, \quad \beta^i \circ \beta^j = 0. \tag{3.3}$$

Now write the vanishing cycle at a degenerate value of the moduli as

$$\nu = \sum_{i=1}^{g} (m_i \alpha_i + n_i \beta^i). \tag{3.4}$$

We now wish to track what happens to cycles in S when we encircle the singular in moduli space that causes ν to vanish. This is dictated by the Picard-Lefschetz formula [69, 70], which tells us that a cycle $\gamma \in \mathrm{H}_1(S, \mathbb{Z})$ shifts as

$$\gamma \mapsto \gamma - (\gamma \circ \nu)\nu. \tag{3.5}$$

Evaluating this map on the basis (α_i, β^i) then gives rise to a monodromy matrix M_ν. For example, in the genus 1 case we have

$$(\alpha, \beta) \mapsto (\alpha, \beta) M_\nu, \quad M_\nu = \begin{pmatrix} 1 - mn & m^2 \\ -n^2 & 1 + mn \end{pmatrix}, \tag{3.6}$$

where we've dropped the subscripts since $g = 1$.

Given the similarities between Picard-Lefschetz theory and the theory of the Coulomb branch, it is interesting to ask if given an $\mathcal{N} = 2$ theory we can find a Riemann surface whose vanishing cycles provide the correct monodromies and homology cycles give rise to good coordinates on the Coulomb branch in some way. This is, in essence, the basis for Seiberg-Witten theory.

Coulomb branch	Seiberg-Witten curve
singular points	vanishing cycles
special coordinates	$\oint_{\alpha_i} \lambda_{\text{SW}}$
dual variables	$\oint_{\beta^i} \lambda_{\text{SW}}$
coupling matrix τ	period matrix BA^{-1}
monodromy	Picard-Lefschetz transformation

Table 3.1. *A dictionary between Coulomb branch data and Seiberg-Witten data.*

3.2 The Seiberg-Witten curve and differential

In [3,4], it was shown that one can indeed rephrase the data of the Coulomb branch in the language of Riemann surfaces. For a theory of rank g, the associated *Seiberg-Witten curve* S is a genus g surface whose vanishing cycles reproduce the correct monodromies of the singular points. However, more data is required in order to extract the special coordinates and their duals.

First note that since S is a surface of genus g, it admits g many holomorphic one-forms ω_i. We can now characterise S by seeing how these one-forms integrate over closed cycles in S. To do so, we define the ***period matrices*** of S to be

$$A_{ij} = \oint_{\alpha_j} \omega_i, \quad B_{ij} = \oint_{\beta^j} \omega_i, \tag{3.7}$$

where α_i and β^i are the symplectic basis of the first homology group discussed previously. Special geometry then allows us to relates these to the special coordinates [71]. Concretely, we have that

$$A_{ij} = \frac{\partial a_i}{\partial u_j}, \quad B_{ij} = \frac{\partial a_i^D}{\partial u_j}, \tag{3.8}$$

where u_i are the moduli of the surface. These equations represent an overdetermined integrability condition that imply the existence of a meromorphic one-form λ_{SW} such that

$$a_i = \oint_{\alpha_i} \lambda_{\text{SW}}, \quad a_i^D = \oint_{\beta^i} \lambda_{\text{SW}}. \tag{3.9}$$

We call λ_{SW} determined in this way the *Seiberg-Witten differential.*

Now we have that a_i and a_i^D in this form, we can see that under a Picard-Lefschetz transformation we have

$$a_i \mapsto a_i - (\alpha_i \circ \nu)\nu, \quad a_i^D \mapsto a_i^D - (\beta^i \circ \nu)\nu \tag{3.10}$$

where ν is the vanishing cycle in question. As such, we see that a_i and a_i^D transform as we would expect from physics. We can also easily determine where on the Coulomb branch these monodromies are sourced. Suppose the Seiberg-Witten curve is written in Weierstraß form

$$y^2 = P(x,u) \equiv \prod_i (x - r_i(u)), \tag{3.11}$$

where $P(x,u)$ is a polynomial in x and the moduli u. For generic values of u, the roots are all distinct and each root can been seen as a branch point when consider the curve as a double cover of $\mathbb{C}$ ramified over these points. Now it is clear if two roots coincide at some value of the moduli, then a cycle will vanish and the Coulomb branch develops a singularity there. As such, the problem of finding singular points of the Coulomb branch amounts to solving the equation

$$\mathrm{Disc}_x(P) = \prod_{i<j} (r_i(u) - r_j(u))^2 = 0. \tag{3.12}$$

Example. Consider pure SU(2) gauge theory. The Seiberg-Witten curve for this theory is given by

$$y^2 = P(x,u) = (x^2 - u^2)^2 - \Lambda^4, \tag{3.13}$$

where u is the modulus and Λ is the energy scale. In this case the discriminant can be easily calculated as

$$\mathrm{Disc}_x(P) = 256\Lambda^8(u^2 - \Lambda^4). \tag{3.14}$$

From this we see that a cycle vanishes when $u^*_\pm = \pm\Lambda^2$. By tracking how the roots of eq. (3.13) move as u approaches $u^*_\pm$, and using eq. (2.28) we can easily see that the singularities correspond to a dyon of charge $(2,-1)$ and a monopole of charge $(0,1)$. The local monodromies are then dictated by eq. (3.6) so

$$M_{(2,-1)} = \begin{pmatrix} 3 & 4 \\ -1 & -1 \end{pmatrix}, \quad M_{(0,1)} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}. \tag{3.15}$$

It is now easy to check that $M_{(2,-1)}M_{(0,1)}$ reproduces the monodromy at infinity given in eq. (2.32). Finally, should we want explicit expressions for (a, a^D) we can obtain them by integrating the Seiberg-Witten differential

$$\lambda_{\mathrm{SW}} = \frac{x^2}{\pi\sqrt{2}} \frac{\mathrm{d}x}{y}, \tag{3.16}$$

over the cycles of the curve. This can be done explicitly using hypergeometric functions [4,72]. In turn, this can be used to calculate $\tau(a)$ explicitly as the complex structure of the torus.

3.3 Non-Lagrangian theories

We have seen that the singular points on the Coulomb branch signal that some BPS states become massless at those values of the moduli and that finding such points is equivalent to finding the roots of the discriminant of the Seiberg-Witten curve. However, if the Seiberg-Witten curve depends on more than one parameter, masses for example, we can also ask how these singularities move while varying these extra parameters. Let us illustrate what can happen with a classic example.

Example. An important example of this occurs in pure SU(3) gauge theory. The Seiberg-Witten curve for this theory is given by

$$y^2 = (x^3 - ux - v)^2 - \Lambda^6 = P(x, u, v). \tag{3.17}$$

The singular points are found by finding the roots of

$$\mathrm{Disc}_x(P) = 64\Lambda^{18}(729\Lambda^{12} + (4u^3 - 27v^2)^2 - 54\Lambda^6(4u^3 + 27v^2)). \tag{3.18}$$

Let us denote this polynomial by Q. If we solved Q for u, we would find the dependence of the roots on v also. An interesting scenario can occur if two roots collide. Physically, this would signal the existence of two simultaneously massless BPS states at the value of this shared root. If the two states in question are ***mutually non-local***, meaning their Dirac pairing is non-zero, then it means that we cannot find a dual frame where only electric degrees of freedom are massless. This is simply due to the fact that the Dirac pairing is invariant under $\mathrm{Sp}(2r, \mathbb{Z})$ duality transformations. As such, this signals the existence of a ***non-Lagrangian theory***, since it is not possible to find a Lagrangian describing such a situation. Returning to the example, this situation can occur if we now look at the discriminant of Q. Doing so gives,

$$\mathrm{Disc}_u(Q) \propto \Lambda^{18} v^6 (\Lambda^{12} - 2\Lambda^6 v^2 + v^4)^2. \tag{3.19}$$

Solving both discriminants simultaneously gives us values of u and v for which mutually non-local matter can become simultaneously massless. One such point is given by $u = 0$ and $v = -\Lambda^3$. By zooming into a neighbourhood of this point, we are left with the non-Lagrangian theory found in [6]. In modern parlance, we refer to this theory as the A_2 ***Argyres-Douglas theory*** and the point in moduli space where is occurs as an ***Argyres-Douglas point.***

It should be noted that Argyres-Douglas points are not unique to the above example. In fact, it was quickly seen that both pure gauge theories and gauge theories with matter can admit such points [6, 73]. In fact, the resulting non-Lagrangian theory found in pure SU(3) gauge theory can also be constructed from an Argyres-Douglas point of SU(2) with one fundamental hypermultiplet [73].

Notice that in decoupling the Argyres-Douglas theory from the $\mathrm{U}(1)^r$ effective theory, we zoomed in locally around the Argyres-Douglas point. As such, the theory is scale invariant by construction and the resulting field theory is conformal. In fact, non-Lagrangian SCFTs make up a large part of the known landscape of higher dimensional SCFTs– most of the known SCFTs in dimensions five and higher are non-Lagrangian.

3.4 Donagi-Witten integrable systems

So far we've only discussed Riemann surfaces and their role in Coulomb branch physics. However, there is a way of rephrasing what we've discussed in the more general framework of abelian varieties. Doing so will give rise to a surprising structure that we can use as an alternative characterisation of the Coulomb branch.

Recall that an ***abelian variety*** A over $\mathbb{C}$ is a quotient of $\mathbb{C}^g$ by a lattice $L \subset \mathbb{C}^g$ of rank $2g$ such that A is a complex projective variety over $\mathbb{C}$. This last condition is equivalent to saying that A admits a holomophic line bundle whose curvature ρ is positive definite after multiplication by i. The $(1,1)$-form $\xi = \mathrm{i}\rho$ is called the ***polarisation*** of A. The polarisation is said to be ***principal*** if

$$\frac{1}{g!}\int_A \xi^g = 1. \tag{3.20}$$

There is a well-known construction that produces a principally polarised abelian variety given an algebraic curve C called the Abel-Jacobi map. The resulting variety is called the ***Jacobian variety*** of C and is constructed as follows.

Take an algebraic curve C of genus g. As discussed earlier, there exists a basis of g many holomorphic one-forms ω_i on C. We define the following $2g$ many vectors

$$\Pi_i = \left(\oint_{\gamma_i} \omega_1, \oint_{\gamma_i} \omega_2, \ldots, \oint_{\gamma_i} \omega_{g-1}, \oint_{\gamma_i} \omega_g\right), \tag{3.21}$$

where γ_i are a basis of $\mathrm{H}_1(C,\mathbb{Z})$. We can now define a lattice L spanned by the Π_i. The Abel-Jacobi map $J : C \to \mathbb{C}^g$ is then given by

$$p \mapsto \left(\int_{p_0}^{p} \omega_1, \int_{p_0}^{p} \omega_2, \ldots, \int_{p_0}^{p} \omega_{g-1}, \int_{p_0}^{p} \omega_g\right) \bmod L, \tag{3.22}$$

where p_0 is some fixed base point[2]. The image of this map is the Jacobian $J(C) \cong \mathbb{C}^g/L$. It now suffices to check whether $J(C)$ admits a polarisation to

[2]Under change of base point, the image of this map is simply translated.

see that it is, indeed, an abelian variety. In fact, it can be shown that $J(C)$ is always a principally polarised abelian variety [74].

Equipped with Abel-Jacobi map, we can now consider the Jacobians of the Seiberg-Witten curves instead of the curves themselves. However, note that the category of polarised abelian varieties is much larger than the category of algebraic curves. Indeed, not all principally polarised abelian varieties arise as the Jacobian of an algebraic curve and the problem of identifying which abelian varieties arise as Jacobians is known as the *Schottky problem*[3].

The reason for introducing abelian varieties is due to their relation with integrability. An ***algebraically integrable system*** is an algebraic manifold X that is the total space of a fibration $\pi : X \to B$ equipped with a closed holomorphic two form σ such that the fibres $\pi^{-1}(b)$ are polarised abelian subvarieties that are Lagrangian with respect to σ [76]. Now consider a theory with Seiberg-Witten curve C. We can construct its Jacobian $J(C)$ and fibre it over the Coulomb branch $\mathcal{B}$. We can then give this fibration a holomorphic two form σ by defining

$$\sigma = \mathrm{d}\lambda_{\mathrm{SW}}, \tag{3.23}$$

and extending it to the total space. It was first noticed in [77] that this constitutes an algebraically integrable system. Furthermore, it was proven in [36] that the total space X of this fibration is always hyper-Kähler and that from any algebraically integrable system one can construct a special Kähler geometry. The integrable systems associated to physical theories are known as ***Donagi-Witten integrable systems*** in the physics literature.

3.5 Scale invariant Coulomb branches

The discussion so far has applied to any $\mathcal{N} = 2$ quantum field theory, but let us now see what restrictions conformal symmetry puts on the special geometry of the theory. This has been extensively studied in rank-1 [78–81] and considerable progress has recently been made for higher ranks [10–16].

Denote the coordinates on the Coulomb branch by u_i for $i = 1, \ldots, r$ with r the rank of the theory. Then the assumption that the Coulomb branch is scale invariant gives us an action of $\mathbb{C}^\times$ on the coordinates as

$$u_i \mapsto \mu^{\Delta(u_i)} u_i, \tag{3.24}$$

where $\Delta(u_i)$ are the scaling dimensions of the coordinates. Geometrically, the invariance under this action implies that the Coulomb branch is metrically a Kähler cone. For rank 1 theories, this means that singularities can only occur at the tip of the cone.

[3]While this problem is largely open, a classic result of Mastusaka tells us that every abelian variety over an algebraically closed field can be written as a quotient of some Jacobian [75].

Kodaira label	Intersection matrix	$\Delta(u)$	Deficit angle
II^*	$\hat{E}_8$	6	$5\pi/3$
III^*	$\hat{E}_7$	4	$3\pi/2$
IV^*	$\hat{E}_6$	3	$4\pi/3$
I_0^*	$\hat{D}_4$	2	π
IV	$\hat{A}_2$	$3/2$	$2\pi/3$
III	$\hat{A}_1$	$4/3$	$\pi/2$
II	$\emptyset$	$6/5$	$\pi/3$
I_n^*	$\hat{D}_{4+n}$	2	2π
I_n	$\hat{A}_{n-1}$	1	2π

Table 3.2. *The Kodaira classification of elliptic fibrations. Listed are the Kodaira labels, intersection matrix of the singular fibre, scaling dimension and deficit angle. The final two are not scale invariant, but can appear as singular fibres for generic quantum field theories.*

We can further restrict the geometry by considering the fact that the Seiberg-Witten curve defines an elliptic fibration over the Coulomb branch in rank 1. As such, the Coulomb branch is an elliptic surface as classified by Kodaira [17]. Kodaira found that there are only nine possible ways for elliptic curve to degenerate at singular points. Imposing the scale invariance property further reduces the possibilities to just seven. Together this has the geometric implication that the cone at the base of the fibration can only have finitely many deficit angles. The full classification is presented in table 3.2.

Even though we have established that there are seven possible rank-1 Coulomb branches, it does not mean that there are just seven rank-1 $\mathcal{N} = 2$ SCFTs. Indeed, several theories can share the same Coulomb branch as there is additional information required to fully specify the theory. One such piece of information is the flavour symmetry that is supported by the singularity. For example, by consulting table 4 in [13], we can see that the II^* singularity can allow for flavour symmetries E_8, $\mathrm{Sp}(10)$ and $\mathrm{SU}(4)$. When factoring this additional information, it turns out that there are remarkably few rank-1 SCFTs– there are approximately 28 up to discrete gauging [78–81].

Moving to higher ranks makes a classification scheme much more difficult as there is no simple analogue of the Kodaira classification. However, there is a useful structure that appears at rank-2 and above that allows us to leverage our knowledge of lower rank theories in order to understand the full theory in question.

Given a rank r theory with $r > 1$, the metric singularities do not occur at isolated points in the Coulomb branch S. Indeed, finding the zeroes of the discriminant of the Seiberg-Witten curve will give the a dependence of the singularities on the other parameters, thus defining a codimension-1 subvariety of the Coulomb branch. These codimension-1 subvarieties $\bar{S}_i^{(r-1)}$ can then intersect to form codimension-2 subvarieties $\bar{S}_j^{(r-2)}$ and so on. This defines a *stratification* of the Coulomb branch [11, 12]

$$\bar{S}^{(0)} \subset \bar{S}^{(1)} \subset \ldots \subset \bar{S}^{(r-1)} \subset S, \tag{3.25}$$

where

$$\bar{S}^{(d)} = \bigsqcup_j \bar{S}_j^{(d)}. \tag{3.26}$$

The key observation here is that in conformal theories, the theories support along the strata are either IR-free or conformal themselves [11, 12]. Furthermore, the strata must still be invariant under the $\mathbb{C}^\times$ action discussed above. This greatly constrains their forms as subvarieties of the Coulomb branch. In particular, $\bar{S}^{(0)}$ must be the origin of S and the full rank r SCFT must be supported here.

Let us restrict now to the case of rank 2 SCFTs and let (u, v) be coordiantes on the Coulomb branch. The $\mathbb{C}^\times$ invariance requires the codimension-1 stratum to have irreducible components of the form

$$\begin{aligned} \mathscr{S}_u : &\quad u = 0, \\ \mathscr{S}_v : &\quad v = 0, \\ \mathscr{S}_{\text{knot.}} : &\quad u^p + \lambda v^q = 0, \end{aligned} \tag{3.27}$$

where p and q are integers fixed by the scaling dimensions of u and v. The presence of the final component, called a *knotted stratum*, is required for the theory to not be trivially decomposable into two rank-1 theories. Furthermore, it can be shown [11, 12] that $\mathscr{S}_u$ is only allowable if $\Delta(v)$ is an allowable rank-1 scaling dimension and vice versa.

The power of this approach is that it can be used to posit the existence of new rank-2 theories and calculate their conformal anomalies. Currently, there have been over 70 rank-2 SCFTs analysed with this approach, giving rise to the current catalogue in [13].

Part II:
Geometric engineering of QFTs

4. Geometric engineering and quivers

4.1 Compactification basics

The core aim of *geometric engineering* is to use the framework of string theory to obtain exact results for quantum field theories [82]. In this thesis, we are mostly interesting in engineering BPS quivers for four dimensional $\mathcal{N} = 2$ theories, so let us briefly review how to obtain such theories through geometric engineering before diving into the specifics.

String theory broadly refers to theories whose fundamental excitations are, as the name suggests, strings and not particles. There are various forms of string theories, including bosonic variants, but the requirement that the spectrum of the theory contains no tachyons restricts us to the so called *superstring theories*.

Superstring theory is a collective term for five different supersymmetric theories in ten spacetime dimensions. While different, the theories are linked by highly non-trivial dualities that allow us to move from one theory to another for computation [83, 84]. Furthermore, these five theories are believed to be different low energy limits of an eleven dimensional theory called *M-theory* [85]. We are only interested in the case of maximally supersymmetric string theories[4] so this reduces us to type IIA, type IIB and M-theory.

The process of obtaining a quantum field theory from a given string theory set-up is called *compactification*. The general idea is to study a given version of string theory on a background $\mathbb{R}^d \times \mathcal{M}$ where $\mathcal{M}$ is manifold such that $d + \dim_{\mathbb{R}} \mathcal{M}$ equals 10 for string theory or 11 for M-theory. In the limit where we take the volume/size of $\mathcal{M}$ to zero, we are left with a theory defined on $\mathbb{R}^{d-1,1}$ that depends on the internal manifold $\mathcal{M}$ in some way. Conversely, demanding that the engineered quantum field theory have certain properties constrains the geometry of $\mathcal{M}$. Note that we can also do this on non-compact manifolds with singularities. Doing so, we will obtain theories with gravity decoupled and we will only require a local understanding of the geometry at the singularities [82].

Since we are concerned in engineering theories with some amount of supersymmetry, let us see what imposing this requires from the geometry of the internal manifold. The requirement that there is some residual supersymmetry after reduction means that there is some vacuum $|0\rangle$ such that

$$\epsilon Q|0\rangle = 0, \quad \langle \delta_\epsilon \Phi \rangle = \langle 0|[\epsilon Q, \Phi]|0\rangle = 0, \tag{4.1}$$

[4] In ten and eleven dimensions this corresponds to 32 supercharges.

$\mathcal{M}$	$\dim_{\mathbb{R}}\mathcal{M}$	$\mathrm{Hol}(\mathcal{M})$	Calibrated cycles
Kähler	$2n$	$\mathrm{U}(n)$	complex
Calabi-Yau	$2n$	$\mathrm{SU}(n)$	special Lagrangian
Hyper-Kähler	$4n$	$\mathrm{Sp}(n)$	—
Quaternionic Kähler	$4n$	$\mathrm{Sp}(n)\cdot\mathrm{Sp}(1)$	—
G_2	7	G_2	(co)-associative
Spin(7)	8	Spin(7)	Cayley

Table 4.1. *The simply connected special holonomy manifolds as classified by Berger [87]. Note that while both Kähler and quaternion Kähler manifolds are included, they are not necessarily Ricci flat.*

for any field Φ. Here Q is the supersymmetry generator and ϵ is spinor parameterising the transformation. In the reduction to d-dimensional Minkowski space, we want to ensure that we preserve the $\mathrm{SO}(d-1,1)$ isometries of the space. This requires that for any field, with the exception of the metric, that is not an $\mathrm{SO}(d-1,1)$ scalar, we have that $\langle\Phi\rangle = 0$. Since spinors are not $\mathrm{SO}(d-1,1)$ scalars, any fermionic field must have vanishing expectation value. In particular, we have that $\langle\delta_\epsilon\Phi\rangle = 0$ for any bosonic field Φ. Now we simply need to check $\langle\delta_\epsilon\Psi\rangle$ for any fermionic fields Ψ in the string theory set up.

Type II string theories possess two fields spin-3/2 in the supergravity approximation called the *gravitini* ψ. Their supersymmetry variation is given by

$$\delta_\epsilon\psi = \nabla_M\epsilon + \dots, \tag{4.2}$$

where ∇_M is the spin covariant derivative of the manifold in question and the additional terms are only dependent on bosonic fields whose expectation values may be taken to be zero in the absence of flux. Taking the expectation value of the above equation tells us that

$$\langle\nabla_M\epsilon\rangle = 0. \tag{4.3}$$

This is equivalent to saying that $\mathcal{M}$ possesses a *covariantly constant* or *parallel spinor*. An immediate consequence of the existence of a parallel spinor is that the manifold $\mathcal{M}$ is necessarily Ricci flat [86]. In fact, the presence of such spinors has much more drastic consequences for the manifolds. This becomes clear when we consider the parallel transport of vectors on $\mathcal{M}$.

Let ∇ be the Levi-Cevita connection on $\mathcal{M}$ and $\mathrm{T}\mathcal{M}$ be its tangent bundle. Furthermore, let $P_\gamma : \mathrm{T}_x\mathcal{M} \to \mathrm{T}_x\mathcal{M}$ be the map capturing how a vector changes upon parallel transport around a contractable loop γ based at x.

Then the *restricted holonomy group* of $\mathcal{M}$ based at $x \in \mathcal{M}$ is defined as

$$\mathrm{Hol}_x(\mathcal{M}) = \{P_\gamma \in \mathrm{GL}(\mathrm{T}_x\mathcal{M}) \,|\, [\gamma] = [0] \in \pi_1(\mathcal{M}, x)\}. \tag{4.4}$$

If $\mathcal{M}$ is simply connected, then the dependence on x can be dropped. Note that the holonomy group of a Riemannian manifold is a subgroup of $\mathrm{O}(2n)$ where $n = \dim_{\mathbb{C}}\mathcal{M}$. However, it was shown in [86] that simply connected Riemannian manifolds admit parallel spinors if and only if their holonomy group is, in fact, a proper subgroup of $\mathrm{U}(n)$. In other words, they are Ricci flat *special holonomy manifolds* as classified by Berger [87]. The classification is presented in table 4.1.

The most famous manifolds of special holonomy type are the *Calabi-Yau* manifolds. Their holonomy groups are $\mathrm{SU}(n)$ and they preserve one quarter of the supersymmetry upon reduction. For example, reducing type II string theory on a Calabi-Yau produces a four dimensional $\mathcal{N} = 2$ theory. From here on in, we will take all internal manifolds to be Calabi-Yau.

4.2 Branes and BPS states

It was realised in the 1980s and 1990s that string theories possess dynamical objects other than strings called *D-branes* [88, 89]. D-branes are extended objects that arise from allowing strings to end with Dirichlet boundary conditions. Traditionally, it was thought that string theory forbade Dirichlet boundary conditions as it would violate Poincaré symmetry. However, the realisation was that strings could end on certain submanifolds specified by the boundary conditions so long as the these submanifolds were dynamically evolving themselves. These submanifolds are exactly the D-branes of string theory. It is typical to call these branes Dp-branes where $p+1$ is the dimension of the submanifold in question.

Crucially, D-branes carry conserved charges which ensure their stability in type II theories. These charges are sources for a $(p+1)$-gauge fields coupled to the Dp-brane (see [90] for an extended discussion). Therefore, the stable branes of type II string theories are dictated by their p-form gauge field content. In type IIA this gives us Dp branes with even p, while type IIB contains Dp branes with odd p[5]. The presence of such stable branes in string theory comes at a cost, however. They only preserve half of the supersymmetries of the theory. As such, they are sometimes referred to as *half BPS branes*[6]. A similar analysis for M-theory shows that it contains three dimensional and six dimensional BPS branes called M2 and M5 branes respectively.

[5] Technically there is also the D(-1) brane in type IIB, but this will be insignificant for us.

[6] Note that there are also stable non-BPS brane configurations [92], but they will not factor into our discussion.

Remark. Since stable D-branes carry charges that are sources for gauge fields of one degree higher, it is natural to ask if these charges are classified by cohomology groups. However, it turns out that this isn't quite the case. In fact, the more sophisticated machinery of K-theory is required to fully classify them [91, 92]. We will not dwell on this, but when phrased in this language the correspondence described in the next section takes on a remarkably simple form.

Now consider type II string theory on a background $\mathbb{R}^d \times \mathcal{M}$ with a Dp-brane wrapping a compact p-cycle C in $\mathcal{M}$. Upon reduction, the engineered QFT sees this as a particle with mass given by

$$M = \mathrm{vol}(C). \tag{4.5}$$

It is clear to see that if C is volume minimising within its own homology class, then the particle in question is in fact a BPS particle. Furthermore, the volume minimising cycles can be described succinctly in terms of *calibrations.*

A Riemannian manifold $\mathcal{M}$ is said to be calibrated if there exists a closed p-form ϕ such that

$$\phi|_{\xi} = \alpha \mathrm{vol}_{\xi}, \quad 0 < \alpha \leq 1, \tag{4.6}$$

for all tangent p-planes ξ [93]. Then it is immediate that any submanifold $\mathcal{N}$ of $\mathcal{M}$ that satisfies

$$\phi|_{\mathcal{N}} = \mathrm{vol}_{\mathcal{N}} \tag{4.7}$$

are volume minimising in their homology class. Such submanfiolds are called *calibrated submanifolds.* Calibrations are well studied in the case of special holonomy manifolds and we present their calibrated submanifolds in table 4.1.

In particular, Calabi-Yau manifolds are calibrated by two different forms– the Kähler 2-form ω and the real part of the holomorphic top form $\mathrm{Re}(\Omega)$. The calibrated submanifolds are then complex submanifolds and special Lagrangian submanifolds respectively. Physically this means that in type IIA D2 branes can wrap complex submanifolds to produce BPS particles while in type IIB D3 branes can wrap special Lagrangians to do the same. As such, the calibrated forms play the role of central charges in string theory. For example, in type IIB the BPS bound becomes

$$M = \mathrm{vol}(C) \geq \left| \int_C \Omega \right| = |Z|. \tag{4.8}$$

Understanding all such C that satisfy the above bound then gives us a positive integral basis of BPS states by wrapping them with D3 branes.

4.3 Singularity theory

The geometric engineering perspective allows us to use the machinery of singularity theory and algebraic geometry in order to understand the engineered theories. Let us now review some of this material.

Firstly, assume that we are working with an internal geometry $\mathcal{M}$ defined by a hypersurface equation $W(x_i) = 0$ with coordinates (x_1, x_2, x_3, x_4). Furthermore, assume that W is quasi-homogeneous

$$W(\lambda^{q_i} x_i) = \lambda W(x_i) \tag{4.9}$$

and has an isolated singularity only at $x_i = 0$. The weights q_i allow us to define a quantity

$$\hat{c} = \sum_{i=1}^{4} (1 - 2q_i). \tag{4.10}$$

Physically, this is the central charge of a two dimensional $\mathcal{N} = 2$ theory that is the IR fixed point of a Landau-Ginzburg model with superpotential W. The quantity $\hat{c}$ takes on geometric importance if we consider the moduli space of the Calabi-Yau $\mathcal{M}$. Indeed, the condition that the singularity appears at a finite distance in the moduli space is equivalent to the bound [46, 94]

$$\hat{c} < 2. \tag{4.11}$$

Note that this generalises to $\hat{c} < d - 1$ for hypersurface equations in $\mathbb{C}^{d+1}$.

It is often useful to consider the possible deformations of the singularity at the origin of $\mathcal{M}$.. These deformations are encoded in the *singularity ring* $\mathcal{R}$ given by

$$\mathcal{R} = \mathbb{C}[x_1, x_2, x_3, x_4]/(\partial W), \tag{4.12}$$

where (∂W) is the ideal generated by the partial derivatives of W. The elements of $\mathcal{R}$ can be written in the form

$$x^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} x_3^{\alpha_3} x_4^{\alpha_4}, \tag{4.13}$$

where α is a multi-index. These elements can then deform W through

$$W \mapsto W + \sum_{\alpha \in \mathcal{R}} g_\alpha x^\alpha. \tag{4.14}$$

Topologically, these deformations can split the singularity into less degenerate singular points. Since $\mathcal{R}$ is understood modulo (∂W), no deformation in $\mathcal{R}$ makes the singularity more degenerate. If a deformation D splits the singularity into only non-degenerate points, then we call $W_\epsilon = W + \epsilon D$ a *Morsification* of W.

For geometries of this form, the holomorphic 3-form Ω has the simple form

$$\Omega = \mathrm{Res}\left[\frac{\prod_i \mathrm{d}x_i}{W}\right], \tag{4.15}$$

where $\mathrm{Res}[\cdot]$ denotes the *Poincaré residue.* This can be easily calculated by eliminating one variable x_j from the numerator and replacing the denominator with $\partial_j W$. Since integrating Ω over a calibrated cycle should give the mass of a BPS state, we get that the mass dimension of Ω must necessarily be one. Using this with the above form gives [46]

$$[x^\alpha] = \frac{2}{2-\hat{c}} \cdot Q_\alpha, \quad [g_\alpha] = \frac{2(1-Q_\alpha)}{2-\hat{c}}, \tag{4.16}$$

where Q_α is the weight of the monomial x^α under the scaling action eq. (4.9). The deformation parameters g_α with $[g_\alpha] > 1$ can be identified with Coulomb branch operators.

Recall from chapter 3, BPS states can be found by looking for vanishing cycles in the Seiberg-Witten curve. This can be uplifted to the Calabi-Yau $\mathcal{M}$, where now we look for vanishing cycles by considering the deformations in $\mathcal{R}$. This can be used to find the BPS quiver of the resulting theory engineered from type IIB as follows.

First we consider a deformation $D \in \mathcal{R}$ that Morsifies the hypersurface $W(x_i) = 0$ and define

$$W_\epsilon(x_i) = W(x_i) + \epsilon D(x_i). \tag{4.17}$$

From this we can define the ***Milnor fibre*** $\bar{X}_p = W_\epsilon^{-1}(p) \cap \bar{B}_\varepsilon$ where B_ε denotes a ball of radius ε. The Milnor fibre is a generically non-singular three complex dimensional manifold which we will analyse to understand the vanishing cycles of the geometry. At particular values of p, $\bar{X}_p$ will develop a non-degenerate singular point. If we call such a point p^* then we can track the behaviour of the cycles as we take a path

$$\tau : [0,1] \to \mathbb{C}, \quad \tau(1) = p^*, \tag{4.18}$$

with $\tau(0) = q$ where $\bar{X}_q$ is non-singular. As we take $s \to 1$, we observe that a cycle vanishes in $\bar{X}_{\tau(s)}$. Note that this is always in the middle homology group $\mathrm{H}_3(\bar{X}_q, \mathbb{Z})$, so this reproduces the intuition from the Seiberg-Witten case.

Now we can repeat the above procedure until exhaustion to find all vanishing cycles in $\mathrm{H}_3(\bar{X}_q, \mathbb{Z})$ and calculate their intersection pairings. We note that the number of vanishing cycles will be equal to the number of possible linearly independent deformations in the singularity $\mu(W) = \dim \mathcal{R}$. We call this number the *Milnor number* of the singularity.

From this information it is now clear how to form the BPS quiver for the engineered theory. We have $\mu(W)$ many nodes with arrows dictated by the intersection pairing between the vanishing cycles. Note that in three complex dimensions, the intersection pairing is anti-symmetric as one would expect from the Dirac pairing.

4.4 Exceptional collections

To end this chapter, we will briefly discuss an alternative way to obtain quivers, this time from type IIA string theory.

Recall that in type IIA we have the D0, D2 and D4 branes and that the calibrated cycles are holomorphic cycles. These BPS branes can form intricate bound states that can, in principal, be difficult to describe. However, in the limit of vanishing string coupling, the branes wrapping holomorphic cycles can be accurately modelled by *B branes* [95]. Such branes are boundary conditions for the topological B-model and have a rich description in terms of *coherent sheaves.*

Definition 4.1. Let X be a space and $\mathcal{O}_X$ its sheaf of holomorphic functions. A *quasi-coherent sheaf* $\mathcal{S}$ is a sheaf of $\mathcal{O}_X$-modules for which there exists an exact sequence

$$\mathcal{O}_X^{\oplus I}(U) \to \mathcal{O}_X^{\oplus J}(U) \to \mathcal{S}(U) \to 1, \tag{4.19}$$

for any open set $U \subset X$. Additionally, $\mathcal{S}$ is called a *coherent sheaf* if it also satisfies the following properties

1. For every open set $U \subset X$, there exists an $n \in \mathbb{N}$ and surjective morphism $\mathcal{O}_X^n(U) \to \mathcal{S}(U)$. We say that $\mathcal{S}$ is of finite type over $\mathcal{O}_X$ if this property holds.
2. For any morphism $\phi : \mathcal{O}_X^n(U) \to \mathcal{S}(U)$ of $\mathcal{O}_X$-modules, the kernel of ϕ is of finite type.

The category of coherent sheaves over X, denoted by $\mathrm{coh}(X)$, is known to be an example of an abelian category [96,97]. That is, it is an additive category that contains all kernels and cokernels of morphisms between objects in $\mathrm{coh}(X)$.

The category of B branes can be motivated as follows [97]. As in type IIA, the category of BPS branes consist of branes wrapping holomorphic cycles. One can also include the possibility of having a vector bundle $\pi : E \to C$ over such a holomorphic cycle C. Such a bundle will help describe possible open string states that can end on the cycle in question. By setting the

B field of string theory to zero, we can assume that this bundle is in fact a holomorphic vector bundle. To any vector bundle, one can associate sheaf to it by considering its sheaf of sections $\Gamma(E, C)$. These sheaves are particularly simple, they are said to be ***locally free***. Such sheaves have the property that for any open subset $U \subset X$, we have that

$$\mathcal{S}(U) \cong \mathcal{O}_X^{\oplus n}(U), \tag{4.20}$$

where n is the rank of vector bundle. As such, it is easy to see that the category of locally free sheaves is in fact a subcategory of $\mathrm{coh}(X)$.

The above construction seems to point towards the category of B branes being the category of locally free sheaves with the morphisms encoding string states stretching between branes. However, in such a set up we have neglected the possibility of having several branes together with strings stretching between them in a composite set up. We must therefore include such set ups in our category of B branes if we wish to be complete. These configurations can be nicely encoded in complexes of sheaves

$$\ldots \to \mathcal{S}_n \to \mathcal{S}_{n+1} \to \mathcal{S}_{n+2} \to \ldots \tag{4.21}$$

where each arrow comes along with a differential d_n which satisfies $\mathrm{d}_{n+1}\,\mathrm{d}_n = 0$. Moving from a category to the category of complexes, up to an appropriate notion of isomorphism, is a standard procedure in homological algebra called moving to the ***derived category***. The caveat with this is that this procedure is only well defined for abelian categories. As such, we should abelianise the category of locally free sheaves in order to move the derived category. It is a standard result that the abelianisation of the category of locally free sheaves is exactly $\mathrm{coh}(X)$. All in all, we have that the category of B branes is given by the derived category[7] of coherent sheaves $\mathbf{D}(\mathrm{coh}(X))$ on X.

The upshot of this reformulation is that if we can find a set of objects that generate $\mathbf{D}(\mathrm{coh}(X))$ in an appropriate sense, we can describe the whole category of BPS states with only information about these objects much in the same way that BPS quivers do. This can be achieved by introducing the notion of an ***exceptional collection***.

An exceptional collection of objects $\{\mathcal{E}_i\}$ of $\mathbf{D}(\mathrm{coh}(X))$ is defined by the property that all objects in $\mathbf{D}(\mathrm{coh}(X))$ can be written as direct sums of the exceptional objects and that

$$\mathrm{Hom}(\mathcal{E}_i, \mathcal{E}_j) = \begin{cases} \mathbb{C}, & \text{if } i = j, \\ 0, & \text{otherwise.} \end{cases} \tag{4.22}$$

Additionally, since $\mathbf{D}(\mathrm{coh}(X))$ is a triangulated category it possesses a suspension functor Σ given by translating the complexes by one step [98]. We

[7] We technically mean the derived bounded category consisting of all finite length complexes, but we will be cavalier with the naming.

further require that

$$\mathrm{Hom}(\mathcal{E}_i, \Sigma^m \mathcal{E}_i) = 0, \tag{4.23}$$

for all non-zero m and that

$$\mathrm{Hom}(\mathcal{E}_j, \Sigma^t \mathcal{E}_i) = 0, \tag{4.24}$$

for $i < j$ and all integers t. Physically, such objects are known as *fractional branes* and are necessarily rigid in the sense that their moduli space is simply a point [99]. From this we can immediately form a quiver with nodes given by exceptional objects and adjacency matrix given by

$$a_{ij} = \dim\Big(\mathrm{Hom}(\mathcal{E}_j, \Sigma \mathcal{E}_i)\Big). \tag{4.25}$$

This definition can be motivated by noting that there is a natural antisymmetric pairing on $\mathbf{D}(\mathrm{coh}(X))$ due to the fact that it is an example of a three Calabi-Yau category [100,101]. Being three Calabi-Yau means that all objects of $\mathbf{D}(\mathrm{coh}(X))$ satisfy a Serre duality of the form

$$\mathrm{Hom}(\mathcal{O}, \Sigma^3 \mathcal{O}') = \mathrm{Hom}\left(\mathrm{Hom}(\mathcal{O}', \mathcal{O}), \mathbb{C}\right). \tag{4.26}$$

With this condition in mind, it is easy to see that the pairing

$$\langle \mathcal{O}, \mathcal{O}' \rangle = \sum_{i=0}^{3} (-1)^n \dim\left(\mathrm{Hom}(\mathcal{O}, \Sigma^n \mathcal{O}')\right), \tag{4.27}$$

is antisymmetric. We can identify this with the Dirac pairing, thus motivating the identification eq. (4.25).

All in all, exceptional collections provide an alternative method of calculating the BPS quivers of theories starting from type IIA string theory or M theory instead of type IIB string theory. We refer readers to [99] for more details of how the details of the standard BPS quiver methodology translate into this framework.

5. The McKay correspondence

In this section we discuss the case of geometric engineering on orbifold singularities. We will start with the classic case of $\mathbb{C}^2/G$ and then outline the extension to higher dimensions.

5.1 Orbifold quiver gauge theory

Given a Calabi-Yau $\mathcal{M}$, we can start asking about BPS branes wrapping submanifolds of $\mathcal{M}$ and the theory that lives on their world volume. The simplest possible case is given by $\mathbb{C}^3$ where a D3-brane can propagate freely. If we allow for a stack of N many D3-branes, the world-volume theory is known to be a simple $\mathrm{U}(N)$ gauge theory. Furthermore, the gauge theory possesses $\mathcal{N} = 4$ supersymmetry and as such enjoys an $\mathrm{SU}(4)_R$ R-symmetry. As such, the field content of the world-volume theory is given by a $\mathrm{U}(n)$ gauge field A which is a singlet under the $\mathrm{SU}(4)_R$, a set of Weyl fermions Ψ in the adjoint representation of $\mathrm{U}(n)$ and the fundamental $\mathbf{4}$ of $\mathrm{SU}(4)_R$ and the bosonic superpartners Φ which transforms under the $\mathbf{6}$ of the R-symmetry and the adjoint of $\mathrm{U}(n)$.

If we now pass to $\mathcal{M} = \mathbb{C} \times \mathbb{C}^2/G$ for some finite group $G \leq \mathrm{SU}(2)$, we can obtain the world-volume theory by gauging G inside the $\mathrm{SU}(4)_R$ R-symmetry of the $\mathbb{C}^3$ theory described above [102]. The resulting theory will have $\mathcal{N} = 2$ supersymmetry and field content given by projecting out the matter content not invariant under G.

In order to understand the effect of this orbifolding procedure, it is helpful to decompose $\mathbb{C}^n$ in accordance to our action of G. Concretely, given the action of G on $\mathbb{C}^n$, we decompose the space as

$$\mathbb{C}^n \cong \bigoplus_i V_i \otimes \mathrm{Hom}_G(V_i, \mathbb{C}^n), \tag{5.1}$$

where V_i are irreducible representations of G and $\mathrm{Hom}_G(V_i, \mathbb{C}^n)$ is the space of G-equivariant homomorphisms from V_i to $\mathbb{C}^n$. We can write the homomorphism spaces as $\mathbb{C}^{N_i}$ for some integers N_i to write this as

$$\mathbb{C}^n \cong \bigoplus_i V_i \otimes \mathbb{C}^{N_i}, \tag{5.2}$$

with the condition that $\sum_i N_i \dim V_i = n$. Note that, by definition, the spaces $\mathbb{C}^{N_i}$ are invariant under G.

First we need to determine what the gauge group is broken to. To do this, we consider the gauge fields A to belong to the space $\mathrm{Hom}(\mathbb{C}^n, \mathbb{C}^n)$ and take the G-invariant part to determine the new gauge group. Doing so, we get

$$\begin{aligned} \mathrm{Hom}_G(\mathbb{C}^n, \mathbb{C}^n) &= \bigoplus_{i,j} \left(\mathbb{C}^{N_i} \otimes (\mathbb{C}^{N_j})^* \otimes V_i \otimes V_j^*\right)^G, \\ &= \bigoplus_i \left(\mathbb{C}^{N_i} \otimes (\mathbb{C}^{N_i})^*\right), \\ &= \bigoplus_i \mathrm{Hom}(\mathbb{C}^{N_i}, \mathbb{C}^{N_i}). \end{aligned} \tag{5.3}$$

where we've used Schur's lemma to get $(V_i \otimes V_j^*)^G = \delta_{ij}$. This tells us that the gauge group $\mathrm{U}(n)$ breaks to $\prod_i \mathrm{U}(N_i)$ after orbifolding.

Similarly, we now look at how the other field content decomposes under the orbifolding. To do so, we now include a non-trivial representation of the R-symmetry in the above calculation. As such, we obtain

$$(\mathcal{V} \otimes \mathrm{Hom}(\mathbb{C}^n, \mathbb{C}^n))^G = \bigoplus_{i,j} a_{ij}^{\mathcal{V}} \left(\mathbb{C}^{N_i} \otimes (\mathbb{C}^{N_j})^*\right), \tag{5.4}$$

where $a_{ij}^{\mathcal{V}}$ is defined through

$$\mathcal{V} \otimes V_i = \bigoplus_j a_{ij}^{\mathcal{V}} V_j. \tag{5.5}$$

This tells us that the Weyl fermions in the $\mathbf{4}$ of $\mathrm{SU}(4)_R$ become a set of bifundamentals for the new gauge group with multiplicities dictated by the matrix $a_{ij}^{\mathbf{4}}$ and similarly for the bosons.

We can easily invert eq. (5.5) by using character theory [103]. Indeed, using the orthogonality of characters we can write

$$a_{ij}^{\mathcal{V}} = \frac{1}{|G|} \sum_k \chi_{\mathcal{V}}(g) \chi_i(g) \overline{\chi_j(g)}. \tag{5.6}$$

Since $a_{ij}^{\mathcal{V}}$ is matrix with non-negative integer entries, we can view is as the adjacency matrix of a directed graph. The resulting graph is called the *McKay graph* or *McKay quiver* relative to $\mathcal{V}$. The field content of the theory is then summarised by the McKay quivers for $\mathcal{V} = \mathbf{4}$ and $\mathcal{V} = \mathbf{6}$.

Given that we are currently interested in the case of $\mathcal{M} = \mathbb{C} \times \mathbb{C}^2/G$, we can choose to decompose $\mathbf{4}$ and $\mathbf{6}$ as

$$\begin{aligned} \mathbf{4} &= \mathbf{1} \oplus \mathbf{1} \oplus 2, \\ \mathbf{6} &= \mathbf{1} \oplus \mathbf{1} \oplus 2 \oplus \overline{2}, \end{aligned} \tag{5.7}$$

where **1** is the trivial representation of G, **2** is a faithful representation of G contained in SU(2) and $\overline{\mathbf{2}}$ is its conjugate representation. Then the world-volume theory is entirely determined by a^2_{ij}. The quivers of this type have long been of interest and are the subject of the *McKay correspondence.*

5.2 The SU(2) McKay correspondence

The McKay correspondence asks what possible McKay quivers one can get by considering finite subgroups of SU(2). Klein showed that any finite subgroup of SU(2) is necessarily falls into one of the following classes [104]:

1. The cycle groups $\mathbb{Z}_p = \langle g : g^q = 1 \rangle$,
2. The binary diherdral groups $\mathbf{BD}_n = \langle r,s,t : r^2 = s^2 = t^n = rst = 1 \rangle$,
3. The binary tetrahedral group $\mathbf{BT} = \langle r,s,t : r^2 = s^3 = t^3 = rst = 1 \rangle$,
4. The binary octahedral group $\mathbf{BO} = \langle r,s,t : r^2 = s^3 = t^4 = rst = 1 \rangle$,
5. The binary icosahedral group $\mathbf{BI} = \langle r,s,t : r^2 = s^3 = t^5 = rst = 1 \rangle$.

Since these groups arise as finite subgroups of SU(2), they each posses a faithful representation whose images are contained in SU(2) which we will call the *canonical representation* of G in SU(2). There are several nice properties of the McKay quivers relative to the canonical representations. For example, it is easy to show that the adjacency matrix of such quivers are symmetric. In this case we drop all orientation on the arrows and simply draw an undirected graph. The most remarkable feature, however, is illustrated by the following theorem of McKay.

Theorem 5.1. (McKay [105]). The finite subgroups of SU(2) are in one-to-one correspondence with the affine ADE Dynkin diagrams through their McKay quivers relative to the canonical representations. Furthermore, the imaginary root in the ADE Dynkin diagram corresponds to the trivial representation.

More concretely, if we were to plot the McKay quivers mentioned in the theorem, we would arrive exactly at those graphs in table 5.1. As such, the theorem also implicitly classifies all possible D3-brane world-volume theories on geometries of the form $\mathcal{M} = \mathbb{C} \times \mathbb{C}^2/G$.

While theorem 5.1 gives a remarkable correspondence between the finite subgroups of SU(2) and the ADE Lie algebras, the correspondence becomes even more remarkable when we consider trying to resolve the geometry $\mathbb{C}^2/G$.

Consider the space $X = \mathbb{C}^2/G$ with G acting through its canonical representation in SU(2). This space is singular at the origin so we can try to find a non-singular variety $\tilde{X}$ which is birationally equivalent to X at generic points of X. Over a field of characteristic zero, one can always do this by

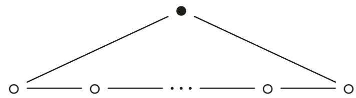
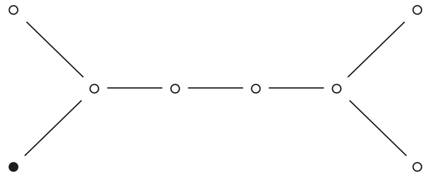
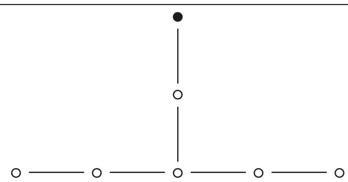
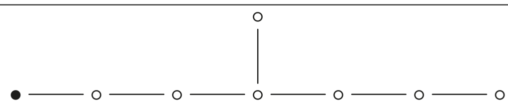
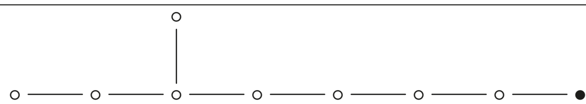
Γ	$\mathfrak{g}_\Gamma$	Affine Dynkin diagram
$\mathbb{Z}_n$	$\mathfrak{su}(n)$	
$\mathbf{BD}_n$	$\mathfrak{so}(2n)$	
BT	$\mathfrak{e}_6$	
BO	$\mathfrak{e}_7$	
BI	$\mathfrak{e}_8$	

Table 5.1. *The McKay correspondence between finite subgroups of* $\mathrm{SU}(2)$ *and the affine Dynkin diagrams of simply laced Lie algebras. The coloured node corresponds to the trivial representation of* Γ.

repeatedly blowing up the singular point [106]. In doing so the singular point is replaced by a collection of Riemann spheres that form the so-called *exceptional divisor* of $\tilde{X}$. If we perform this for all finite subgroups of SU(2) we arrive at the following theorem.

Theorem 5.2. (Gonzalez-Sprinberg & Verdier [107]). Let $\Gamma \leq \mathrm{SU}(2)$ be a finite group and $X = \mathbb{C}^2/\Gamma$ be the corresponding orbifold. There exists a crepant resolution $f : \tilde{X} \to X$ such that the exceptional divisor of $\tilde{X}$ is a collection of Riemann spheres E_i with intersection numbers

$$E_i \cdot E_j = -C^{\mathfrak{g}}_{ij}, \tag{5.8}$$

where $C^{\mathfrak{g}}$ is the Cartan matrix of the Lie algebra $\mathfrak{g}_\Gamma$ as in table 5.1.

Remark. Note that orbifold singularities have a unique crepant resolution in complex dimension two. This is not the case in higher dimensions. In three dimensions the crepant resolutions are related by flops, while orbifold singularities in dimensions four and higher need not have crepant resolutions.

5.3 Generalisations

We now wish to analyse the case of $\mathcal{M} = \mathbb{C}^3/G$ for $G \leq \mathrm{SU}(3)$. As the McKay correspondence played a key role in the SU(2) case, it is natural to ask if there is a generalisation to SU(3). Here we will outline two such generalisations.

Before doing so, we should note that the finite subgroups of SU(3) were classified in 1905 by Blichfeldt [108] and refined over the years [109]. Appendix A of paper II gives a concise account of these groups, together with presentations for the groups. We also recommend [110] for an accessible account of these groups and their properties for physicists.

5.3.1 The Ito-Reid version

The first version of the generalisation is much like the regular McKay correspondence in spirit, but instead places more emphasis on the conjugacy classes rather than the representations themselves.

Conjecture 5.1. (Ito-Reid [111]). Let $G \leq \mathrm{SU}(n)$ be a finite group, $X = \mathbb{C}^n/G$ be the corresponding orbifold and $f : \tilde{X} \to X$ be a crepant resolution. Then there exists a basis of $\mathrm{H}_*(\tilde{X}, \mathbb{Q})$ consisting of algebraic cycles in one-to-one correspondence with conjugacy classes of G.

Remark. The Ito-Reid version of the McKay correspondence is proven for subgroups of SU(3) through explicit computation (see [112], for example). However, it is unproven for quotients of $\mathbb{C}^4$ and above.

The Ito-Reid version also gives us the tools to tell which conjugacy classes correspond to certain dimensional cycles. For simplicity, let us outline this for $n = 3$ where the conjecture is proven.

Any element $g \in G \subset \mathrm{SU}(3)$ has a finite order r and eigenvalues of the form

$$\boldsymbol{\lambda} = \left[\exp\left(\frac{2\pi \mathrm{i}a}{r}\right), \exp\left(\frac{2\pi \mathrm{i}b}{r}\right), \exp\left(\frac{2\pi \mathrm{i}c}{r}\right)\right] \tag{5.9}$$

where $0 \leq a, b, c < r$. These eigenvalues are clearly shared by all other elements of the conjugacy class of g. We now define the *age* of the conjugacy class to be

$$\mathrm{age}(g) = \frac{1}{r}(a + b + c). \tag{5.10}$$

Note that since G is a subgroup of SU(3), this is always an integer. The Ito-Reid correspondence then says that

$$b_{2m}(\tilde{X}) = |\mathbf{G}_m|, \tag{5.11}$$

where $b_n(\tilde{X})$ is the n^{th}-Betti number of $\tilde{X}$ and $\mathbf{G}_m$ is the set of age m conjugacy classes.

On the physics side, if we engineer a theory from M-theory on a geometry $\mathbb{R}^{4,1} \times \mathbb{C}^3/G$, then the rank ρ and flavour rank f of the theory are determined as [113]

$$\rho = |\mathbf{G}_2|, \quad f = |\mathbf{G}_1| - |\mathbf{G}_2|. \tag{5.12}$$

In [114], an algorithm is presented to find exactly which conjugacy classes correspond to flavour curves, but this will not be important for us.

Example. Consider the abelian group $\mathbb{Z}_4$ with the representation

$$\mathcal{R}(g) = \mathrm{diag}(\mathrm{i}, \mathrm{i}, -1), \tag{5.13}$$

where g generates the $\mathbb{Z}_4$. In this case we have that

$$\mathrm{age}(g) = 1, \quad \mathrm{age}(g^2) = 1, \quad \mathrm{age}(g^3) = 2. \tag{5.14}$$

As such, we have that $b_2(\tilde{X}) = 2$ and $b_4(\tilde{X}) = 1$ together with the physical data $\rho = 1$ and $f = 1$. The corresponding McKay quiver is plotted in fig. 5.1.

Finally, as we mentioned before, the Ito-Reid correspondence puts particular emphasis on the conjugacy classes of G and not the representations.

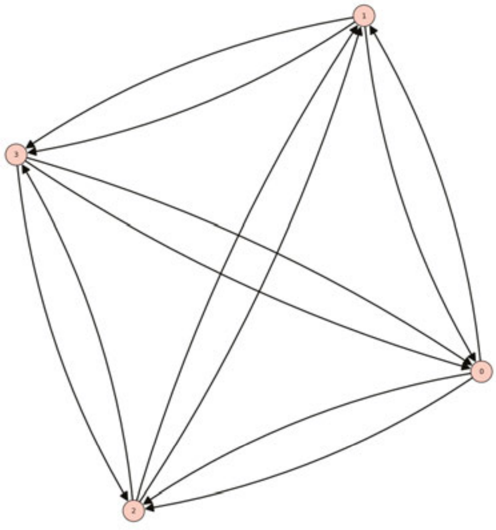


Figure 5.1. The McKay quiver for $\mathbb{Z}_4$ relative to the representation given in the example. Note that the number of nodes is given by $2\rho + f + 1$ where ρ is the rank of the theory and f is the flavour rank.

While the number of both match, it is important to note that the representation ring $\mathrm{Rep}(G)$ and the ring of conjugacy classes $Z(\mathbb{C}[G])$ are not isomorphic. As such, it is not easy to translate the information from the conjugacy classes to the representations, which is how we calculate the McKay quiver. It is therefore useful to have some other generalisation for which information can be more easily interpreted from the quiver.

5.3.2 The McKay correspondence as a derived equivalence

We saw in section 4.4 that one can obtain quivers by considering sheaves instead of simply the homology of the internal geometry. As such, it is not surprising that the McKay correspondence has a variant which places the emphasis on coherent sheaves instead.

Theorem 5.3. (Bridgeland-King-Reid [115]). Let $G \leq \mathrm{SU}(3)$ be a finite group and $X = \mathbb{C}^3/G$. Then there exists a crepant resolution $f : \tilde{X} \to X$ such that the derived category of coherent sheaves $\mathbf{D}(\tilde{X})$ on $\tilde{X}$ is equivalent to the derived category of G-equivariant coherent sheaves $\mathbf{D}^G(\mathbb{C}^3)$ on $\mathbb{C}^3$.

An immediate corollary to this result is that G-equivariant K-theory group $K_0^G(\mathbb{C}^3)$ is isomorphic, as a ring, to the representation ring $\mathrm{Rep}(G)$ of G. Since K-theory groups are known to classify stable D-brane configurations, this matches what we would expect from the point of view of the quiver.

Remark. Note that the result of Bridgeland-King-Reid is much more general than as stated above. In fact, it gives a more concrete realisation of the crepant resolution $\tilde{X}$ in terms of Hilbert schemes.

Part III:
Generalised symmetries and their consequences

6. Symmetries in quantum field theory

6.1 Ordinary symmetries and Noether's theorem

Suppose we are given a theory with Lagrangian $\mathcal{L}$ depending on fields $\{\phi_a\}$. Now consider an infinitesimal transformation of the fields given by

$$\delta\phi_a = X_a(\phi), \tag{6.1}$$

which generates some Lie group G. Under this transformation, the Lagrangian varies as

$$\begin{aligned} \delta\mathcal{L} &= \frac{\partial\mathcal{L}}{\partial\phi_a}\delta\phi_a + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\partial_\mu(\delta\phi_a), \\ &= \left[\frac{\partial\mathcal{L}}{\partial\phi_a} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\right)\right]\delta\phi_a + \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\delta\phi_a\right). \end{aligned} \tag{6.2}$$

By imposing the equations of motion the first term vanishes. Recall that the Lagrangian is defined up to a shift by a total derivative $\delta\mathcal{L} = \partial_\mu F^\mu$. As such, the transformation defines a symmetry if we have

$$\partial_\mu j^\mu := \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}X_a(\phi) - F^\mu\right) = 0, \tag{6.3}$$

for some function F^μ. The function j^μ is called the *conserved current* associated to the symmetry and is often written as a one-form j. From this, we can define a charge operator

$$Q(M^{(d-1)}) = \int_{M^{(d-1)}} \star j, \tag{6.4}$$

where $\star$ denotes the Hodge star operation and $M^{(d-1)}$ is some $(d-1)$-dimensional manifold. Typically, $M^{(d-1)}$ is given by a fixed time slice of spacetime, but this more general definition will be useful. Note that Q will vanish unless we insert charged objects into our theory.

Given this description, several questions immediately come to mind. First of all, how can we deal with discrete symmetries? The construction above relied on a continuous transformation of the fields, so it isn't clear if there is an analogous construction for discrete symmetries. Furthermore, what if a theory is non-Lagrangian? We know that they still possess some notion of symmetry, but how can we describe them?

A more uniform approach is given by recasting the symmetries in terms of topological operators [116]. This can be done generally by cutting spacetime along the manifold $M^{(d-1)}$ and inserting a group transformation on the complete set of states for the Hilbert space associated to this operator. In the continuous case, the operators that implement this are schematically given by

$$U_g(M^{(d-1)}) = \exp\Big(\mathrm{i}gQ(M^{(d-1)})\Big). \tag{6.5}$$

By construction, regardless of whether we are in the continuous or discrete case, we have that these operators satisfy a group law

$$U_g(M^{(d-1)})U_{g'}(M^{(d-1)}) = U_{g\cdot g'}(M^{(d-1)}), \tag{6.6}$$

so the map $g \mapsto U_g(\mathcal{M})$ is a homomorphism for fixed $\mathcal{M}$. The topological property of these operators is clear in the continuous case. Indeed, the conservation of the current j means that the operator is unchanged under small deformations of $M^{(d-1)}$ by Stokes' theorem.

Using the topological operators U_g, we can probe the charges of point-like operators under this symmetry. Concretely, let $\mathcal{V}(p)$ be an operator supported at the point p and take S_p^{d-1} to be a $(d-1)$-dimensional sphere surrounding the point p. Then the action of $U_g(S_p^{d-1})$ on $\mathcal{V}(p)$ is given by

$$U_g(S_p^{d-1})\mathcal{V}(p) = \mathcal{R}(g)\mathcal{V}(p), \tag{6.7}$$

where $\mathcal{R}$ is a representation of the symmetry group. Topologically, we can think about the sphere shrinking around the point p to act upon the operator. The representation $\mathcal{R}$ then encodes the charge of $\mathcal{V}(p)$ under the symmetry group G. If we instead consider $M^{(d-1)}$ to be a fixed time slice, we can write the equal time commutator

$$U_g(M^{(d-1)})\mathcal{V}(p) = \mathcal{R}(g)\mathcal{V}(p)U_g(M^{(d-1)}). \tag{6.8}$$

The non-commutativity arises from the fact that $\mathcal{V}$ is supported on a point in $M^{(d-1)}$ at that fixed time.

6.2 Generalisations

Given a symmetry, we can characterise it by looking at the codimension-1 topological operators that encode it. Interestingly, if we relax some of the conditions of the above construction, we can find new forms of symmetries. In doing so we take the set of topological operators to define the symmetry and put no emphasis on transformations of fields. Here will we note three possible generalisations.

6.2.1 Higher form symmetries

For an ordinary symmetry, the operators $U_g(M)$ are supported on codimension 1 manifolds. The objects charged under such symmetries are operators supported on zero dimensional points whose charge can be measured by taking M to link with the point. Suppose now we relax the condition that M is codimension-1 and consider the set of topological operators to be supported on codimension-p manifolds with $p > 1$. What would this mean for the charged content?

An immediate topological consequence is that operators supported on points would no longer be charged under the symmetry. This is because we can always move the operator away from the point-like operator and shrink it. The action would therefore be trivial and the operator would be uncharged. Instead, we would need charged objects that we could not always do this process with. This means that the charged objects would be able to *link* with the symmetry operators $U_g(M)$. In particular, we should have that

$$\dim M + \dim C = d - 1, \tag{6.9}$$

where C is the support of the operator and d is the spacetime dimension. Now let $\mathcal{V}(C^{(p)})$ be an operator charged under such a symmetry. The generalisation of eq. (6.4) is then given by

$$U_g(\mathcal{S}^{d-p-1})\mathcal{V}(C^{(p)}) = \mathcal{R}(g)\mathcal{V}(C^{(p)}), \tag{6.10}$$

where is $\mathcal{S}^{d-p-1}$ is a small manifold linking $C^{(p)}$. Similarly the equal time commutation relation becomes

$$U_g(M^{(d-p-1)})\mathcal{V}(C^{(p)}) = \mathcal{R}(g)^{\ell(C^{(p)},M^{(d-p-1)})}\mathcal{V}(C^{(p)})U_g(M^{(d-p-1)}), \tag{6.11}$$

where $\ell(\cdot,\cdot)$ denotes the link pairing of the two manifolds.

The group law eq. (6.6) still holds for these operators, but increasing the codimension of these operators has consequences for the symmetry group. In particular, we imagine two operators U_g and $U_{g'}$ acting at times t and $t+\epsilon$. Since these operators are supported on manifolds of at least codimension two, there is no obstruction to deforming the support of $U_{g'}$ so the operator acts at a time $t-\epsilon$. As such, any symmetry implemented by these topological operators must form an abelian group.

Returning to the continuous case, it is clear how to generalise eq. (6.4). Indeed, now have a closed $(d-p-1)$-form $\star J$ integrated over a codimension-$(p+1)$ manifold $M^{(d-p-1)}$. Explicitly, the charge operator is now

$$Q(M^{(d-p-1)}) = \int_{M^{(d-p-1)}} \star J, \tag{6.12}$$

and the topological operators U_g are formed by exponentiating this. The $(p+1)$-form J is now the analogue of the 1-form conserved current j in the ordinary case. As such, we call this type of symmetry a *higher form symmetry* to denote that there is a $(p+1)$-form conserved current and that the objects charged under such a symmetry are support on p-dimensional manifolds.

Example. Consider four dimensional U(1) gauge theory. The action is given by

$$S_{\mathrm{U}(1)} = \frac{1}{g^2}\int F\wedge\star F, \tag{6.13}$$

where $F = \mathrm{d}A$ is the two-form field strength and g is the gauge coupling constant. The field strength is clearly closed, while the equations of motion ensure that $\mathrm{d}\star F = 0$. Interpreting these as conserved currents, we get two sets of U(1) topological operators supported on manifolds of dimension two

$$U_\alpha(M^{(2)}) = \exp\left(\frac{\mathrm{i}\alpha}{g^2}\int_{M^{(2)}}\star F\right),\ V_\alpha(M^{(2)}) = \exp\left(\frac{\mathrm{i}\alpha}{2\pi}\int_{M^{(2)}}F\right) \tag{6.14}$$

where $\alpha\in\mathbb{R}/\mathbb{Z}$. Since the integral over $\star F$ is the simply the electric flux through the manifold $M^{(2)}$, the set of operators U_α generate the *electric 1-form symmetry* of the theory. Concretely, it implements a shift of A to $A+\lambda$ where λ is a flat, but not necessarily exact, connection. This can be easily seen by coupling the theory to a background field B for this symmetry through

$$\tilde{S}_{\mathrm{U}(1)} = \frac{1}{g^2}\int (F-B)\wedge\star(F-B), \tag{6.15}$$

and noticing that shifting $A\mapsto A+\lambda$, with λ not necessarily flat, requires $B\mapsto B+\mathrm{d}\lambda$. Finding the current as usual reproduces the topological operators U_α. The objects charged under the electric 1-form symmetry are the well known Wilson lines.

Similarly, the operators V_α generate the *magnetic 1-form symmetry*. If one also introduces a background field for this symmetry, it is easy to see that there is a mixed 't Hooft anomaly between the two 1-form symmetries. The objects charged under the magnetic 1-form are 't Hooft lines [117].

Now consider adding matter to the above system. The matter will carry a representation of U(1) which defines its electric charge q. The matter then acts as a source in the equations of motion, hence partially breaking the electric 1-form symmetry. This can be easily seen from the operators, since the presence of the source spoils the topological nature of them. However, if one restricts to shifting by flat connections with $\mathbb{Z}_q$ holonomy, then the symmetry remains and the U(1) 1-form symmetry breaks to $\mathbb{Z}_q$. On the other hand, the Bianchi identity always ensures that the magnetic 1-form symmetry operators are topological, so it remains unbroken. In essence, the presence of the charge q matter can screen the Wilson lines, so that the charge modulo q is only important.

6.2.2 Higher group symmetries

The possibility of having higher form symmetries in a system also raises a question: can such symmetries mix with ordinary (0-form) symmetries to enhance the overall symmetry group to something larger?

The correct language to describe this is that of *higher groups* and, in particular, 2-groups. The latter are a categorification of the concept of a regular group and were discussed in various contexts in [118–120]. There are various ways to describe a 2-group, we will follow the discussion given in appendix A of [121].

Consider a theory with continuous 0-form symmetry group G and discrete 1-form symmetry group A. This theory will posses of a spectrum of line operators $\mathscr{L}$. As expected, one can obtain A from $\mathscr{L}$ by defining an equivalence relation on $\mathscr{L}$. We identify two lines $L_1 \sim L_2$ if there exists a point-like *line changing operator* between them. A line changing operator is simply a point-like operator that lies at the junction between lines. If a line can end on such an operator with no other lines present, then it is completely screened by the operator. We therefore define

$$\hat{A} = \mathscr{L}/\sim \tag{6.16}$$

where $\hat{A}$ denotes the Pontryagin dual $\mathrm{Hom}(A, \mathbb{C}^\times)$ of A. While this gives us the 1-form symmetry group, it doesn't quite give us the interplay between the 0-form and 1-form symmetries. This is due to the fact that there could exist line changing operators which do not lie in a linear representation of G. We can therefore refine the equivalence relation $\sim$ by only considering line changing operators which are consistently acted upon by the 0-form symmetry. Denoting this equivalence relation as $\sim'$, we then define the group

$$\hat{A}' = \mathscr{L}/\sim' \tag{6.17}$$

which fits into a short exact sequence

$$1 \to \hat{C} \to \hat{A}' \to \hat{A} \to 1. \tag{6.18}$$

The lines that belong to $\hat{C}$ are able to be screened by point-like operators that lie in a *projective representation* of G. As such, they define a central extension of G given by

$$1 \to C \to \tilde{G} \to G \to 1. \tag{6.19}$$

By dualising eq. (6.18), we can then combine these two short exact sequences to get

$$1 \to A \to A' \to \tilde{G} \to G \to 1, \tag{6.20}$$

which captures the mixing between the 0-form and 1-form symmetries.

Finally, note that eq. (6.19) is captured by an element $\omega \in \mathrm{H}^2(G,C)$ which we can extend to an element $\beta = \mathrm{Bock}(\omega) \in \mathrm{H}^3(G,A)$ where Bock is the Bockstein homomorphism[8] associated with the sequence dual to eq. (6.18) [122, 123]. If β is non-trivial then the 2-group extension captured by the above construction is non-trivial.

All in all, a 2-group can be defined as a quadruple (G,A,α,β) where $\beta \in \mathrm{H}^3(G,A)$ and $\alpha : G \to \mathrm{Aut}(A)$ is an action[9] of G on A. The map α gives the explicit realisation of the action of the 0-form symmetry on the 1-form symmetry. This presentation of a (strict) 2-group is known as a *crossed module* [125].

Here we have simply outlined the structure that appears when a 0-form and a 1-form mix, but it is also known that, in principle, more than two symmetries can mix to form a higher group which can be understood in the language of higher categories or n-crossed modules (see [126] for an example).

6.2.3 Non-invertible symmetries

The final type of generalisation arises when we relax the group law of the topological operators, resulting in the so-called *non-invertible symmetries*. These will not be important for our discussion, but for completeness we give a brief description of them.

In general, we can view the topological symmetry operators as belonging to some *fusion category*[10] $\mathscr{C}$, whose fusion product $\otimes$ may or may not be group-like. A simple class of examples of non-group-like fusion categories are provided by the *Tambara-Yamagami categories* [128] which have been studied extensively in the context of 2d theories [129]. The Tambara-Yamagami category $\mathrm{TY}(G)$ associated to a finite group G is a category whose simple objects are indexed by group elements of G together with an additional object U_m. Their fusion rules are then given by

$$\begin{aligned} U_g \otimes U_{g'} &= U_{gg'}, \\ U_g \otimes U_m &= U_g = U_m \otimes U_g, \\ U_m \otimes U_m &= \sum_{g\in G} U_g. \end{aligned} \tag{6.21}$$

In the fusion rules, we explicitly see that U_m has no inverse in the $\mathrm{TY}(G)$.

[8] The Bockstein homomorphism is just a fancy name for the connecting morphism in the long exact sequence in cohomology associated to a short exact sequence.

[9] There are some mild conditions on this action. We refer readers to [124] for an in depth discussion of the crossed module construction.

[10] More descriptively, a fusion category is a rigid semi-simple linear monoidal category [127].

Until recently, almost all examples of non-invertible symmetries were in two dimensional systems. However, it is now known that non-invertible symmetries are not only possible in higher dimensions, but are in fact quite common [130–137].

6.3 Higher form symmetries and geometric engineering

6.3.1 The defect group

As the charged objects of higher form symmetries are extended operators, we can ask how these operators arise from geometric engineering.

Suppose we wish to engineer an object of dimensionality q from string or M-theory. We've already seen that we can obtain BPS particles by wrapping a $(q+1)$-dimensional brane on a q-dimensional cycle in the internal geometry. It is therefore clear that in order to engineer extended objects, we only partially wrap the brane in the internal geometry and allow a certain number of dimensions to propagate in Minkowski space. Furthermore, since the volume of the wrapped cycle is proportional to the mass of the engineered object, it is clear that wrapping a non-compact cycle in the internal geometry engineers infinitely massive defects. As such, if we wish to capture the spectrum of defects of a theory geometrically, we need to find an object which encodes the non-compact cycles of the internal geometry $\mathcal{M}$. This can be achieved by defining the *defect group* $\mathbb{D}$ to be [21]

$$\mathbb{D} = \bigoplus_{n} \mathbb{D}[n], \quad \mathbb{D}[n] = \bigoplus_{(k,p)\in I_n} \mathrm{Tor}\left(\frac{\mathrm{H}_k(\mathcal{M}, \partial\mathcal{M})}{\mathrm{H}_k(\mathcal{M})}\right), \tag{6.22}$$

where I_n is an index set encoding the possible p-branes and k-cycles such that $p+1-k=n$. Here $\mathrm{H}_\bullet(\mathcal{M}, \partial\mathcal{M})$ denotes the the relative homology of $\mathcal{M}$ with respect to its boundary at infinity. Physically, the defect group encodes the charges of possible defects of an engineered theory and the process of taking the quotient by the compact homology implements a generalisation of 't Hooft screening [138].

Let us now specialise to the case of M theory on $\mathcal{N}\times\mathcal{M}$ where $\mathcal{M}$ is a non-compact Calabi-Yau threefold that can be obtained as the cone over some Sasaki-Einsten manifold Y_5 and $\mathcal{N}$ is a closed spin manifold which we take to be spacetime. Furthermore, we will assume that both $\mathrm{H}_{2n+1}(\mathcal{M})$ and $\mathrm{Tor}(\mathrm{H}_{2n}(\mathcal{M}))$ are trivial for all n. Since the only stable branes of M theory are the M2 and M5, it is easy to see that the defect group will have the form

$$\mathbb{D} = \mathbb{D}[0]\oplus\mathbb{D}[1]\oplus\mathbb{D}[2]\oplus\mathbb{D}[4]. \tag{6.23}$$

Here $\mathbb{D}[0]$ signals a possible (-1)-form symmetry. We will postpone a discussion of this until the next section. Additionally, due to our assumptions, the only possible homology groups with torsion are given by

$$\mathscr{D} = \operatorname{Tor} \mathrm{H}_1(Y_5) = \operatorname{Tor} \mathrm{H}_3(Y_5), \tag{6.24}$$

where the final equality follows from Poincaré duality and the universal coefficient theorem. From this we see that $\mathbb{D}[k] = \mathscr{D}$ for the $k \in \mathbb{Z}$ appearing in eq. (6.23) by comparing the definition of the defect group with the long exact sequence in relative homology for $\mathcal{M}$ and $\partial\mathcal{M}$.

In this set up, there is a simple method of computing the required homology groups. In particular, consider taking taking some crepant resolution of $\mathcal{M}$ and let X_6 be the intersection of this resolution with some sufficiently large ball B_6. Now consider the long exact sequence in relative homology [122]

$$\ldots \to \mathrm{H}_n(Y_5) \to \mathrm{H}_n(X_6) \to \mathrm{H}_n(X_6, Y_5) \to \mathrm{H}_{n-1}(Y_5) \to \ldots \tag{6.25}$$

Assuming that $\mathrm{H}_{2n+1}(X_6)$ is trivial, this sequence will truncate into several smaller sequences. Furthermore, if we assume that X_6 is torsion free, then the universal coefficient theorem and Lefschetz duality allows us to rewrite several of these terms as

$$\mathrm{H}_n(X_6, Y_5) \cong \operatorname{Hom}(H_{6-n}(X_6, Y_5), \mathbb{Z}). \tag{6.26}$$

All in all, we arrive at an exact sequence of the form

$$1 \to \mathrm{H}_2(Y_5) \to \mathrm{H}_2(X_6) \xrightarrow{Q_2} \operatorname{Hom}(\mathrm{H}_4(X_6, Y_5), \mathbb{Z}) \to \mathrm{H}_1(Y_5) \to 1, \tag{6.27}$$

with a similar one given by

$$1 \to \mathrm{H}_4(Y_5) \to \mathrm{H}_4(X_6) \xrightarrow{Q_4} \operatorname{Hom}(\mathrm{H}_2(X_6, Y_5), \mathbb{Z}) \to \mathrm{H}_3(Y_5) \to 1. \tag{6.28}$$

This allows us to make the following identifications

$$\mathrm{H}_1(Y_5) = \operatorname{coker}(Q_2), \quad \mathrm{H}_3(Y_5) = \operatorname{coker}(Q_4). \tag{6.29}$$

Crucially, the maps Q_k have a simple geometric interpretation. They are partial evaluations of the intersection pairings

$$q_k : \mathrm{H}_k(X_6) \to \mathrm{H}_{6-k}(X_6). \tag{6.30}$$

One can now ask if the relation between intersection pairings and the defect group can be used when geometrically engineering theories from string theory instead. As discussed in [139], this relation persists to type IIB on hypersurface singularities. In particular, by doing a similar calculation to as above, we get that the defect group is given by

$$\mathbb{D}_{\mathrm{IIB}} = \mathbb{D}[1] = \operatorname{Tor} \mathrm{H}_2(Y_5) = \operatorname{Tor}(\operatorname{coker}(Q_3)). \tag{6.31}$$

As discussed in section 4.3, we know that the intersection pairing of the internal geometry can be interpreted as the intersection matrix of the BPS quiver for the engineered theory.

6.3.2 Methods of calculation

As we've seen, in the cases discussed above, we can find defect groups by calculating the cokernels of intersection pairings or by, equivalently, finding the homology of the boundary geometry. In this section we will discuss two methods of calculating the required cokernels and homology.

Smith normal form. By giving the appropriate homology groups a basis, we can give Q_k a matrix form B_k. We can regard B_k as a map from a lattice $\mathbb{Z}^D$ to itself. In three complex dimensions B_k is antisymmetric, so it can always be brought into the form

$$B' = PAP^T = \left[\begin{array}{ccccccc|c} 0 & a_1 & 0 & 0 & \cdots & 0 & 0 & \\ -a_1 & 0 & 0 & 0 & \cdots & 0 & 0 & \\ 0 & 0 & 0 & a_2 & \cdots & 0 & 0 & \\ 0 & 0 & -a_2 & 0 & \cdots & 0 & 0 & \mathbf{0}_{2r\times 1} \\ \vdots & \vdots & \vdots & \ddots & \ddots & \vdots & \vdots & \\ 0 & 0 & 0 & 0 & \cdots & 0 & a_r & \\ 0 & 0 & 0 & 0 & \cdots & -a_r & 0 & \\ \hline \multicolumn{7}{c|}{\mathbf{0}_{1\times 2r}} & \mathbf{0}_{f\times f} \end{array}\right] \tag{6.32}$$

where $D = 2r + f$ and P is a unitary matrix. Furthermore, the a_i can be ordered such that $a_{i+1}|a_i$ for all i. It is a standard result in the theory of principal ideal domains that there exist $\mathbb{Z}$-valued matrices S and T such that

$$SB'T = \mathrm{diag}(a_1, a_1, a_2, a_2, \ldots, a_r, a_r, 0, \ldots, 0). \tag{6.33}$$

This is called the *Smith normal form* of B. Given the Smith normal form of B_k, the cokernel of Q_k is given by

$$\mathrm{coker}(Q_k) \cong \mathbb{Z}^f \times \prod_{i=1}^{r} \mathbb{Z}^2_{a_{r_i}}. \tag{6.34}$$

While we usually discard the $\mathbb{Z}$ factors in the definition of the defect group, it still has physical meaning. For example, in the type IIB hypersurface set up, the integer f matches the flavour rank of the engineered theory.

Orlik's algorithm. The Smith normal form method relies on knowing the intersection pairing explicitly. However, this can be difficult in practice. For the case of quasi-homogeneous hypersurfaces, there is a method of obtaining the defect group without calculating the intersection matrix explicitly.

Given a quasi-homogeneous hypersurface X_6 with an isolated singularity, Orlik's algorithm calculates the homology of the *link* $\mathcal{Y} = X_6 \cap S^5_\epsilon$ where S^5_ϵ is a small sphere centred on the singularity. In the case where X_6 is a cone over a manifold Y_5, the link is homeomorphic to Y_5, so we can apply this method in the cases discussed above.

The first step of the algorithm is to find the integer weights and degree of the hypersurface defined by the vanishing locus of $f : \mathbb{C}^4 \to \mathbb{C}$. We define them as

$$f(\lambda^{w_i} x_i) = \lambda^d f(x_i). \tag{6.35}$$

From these numbers we define

$$\begin{aligned} r_1 &= \gcd(w_2, w_3, w_4), \quad r_2 = \gcd(w_1, w_3, w_4), \\ r_3 &= \gcd(w_1, w_2, w_4), \quad r_4 = \gcd(w_1, w_2, w_3), \end{aligned} \tag{6.36}$$

and

$$2g_i = -1 + \sum_{j \neq i} \frac{\gcd(w_i, w_j)}{w_j} + \sum_{\substack{j<k, \\ j,k \neq i}} \frac{\gcd(w_j, w_k)}{w_j w_k} + \frac{d^2 r_i w_i}{w_1 w_2 w_3 w_4}. \tag{6.37}$$

Then Orlik's conjecture [140, 141] states that the torsional part of $\mathrm{H}_2(Y_5)$ is given by

$$\operatorname{Tor} \mathrm{H}_2(Y_5) = \prod_{i=1}^{4} \mathbb{Z}_{r_i}^{2g_i}. \tag{6.38}$$

From the above discussion, this is exactly $\mathbb{D}[1]$ for the engineered theory.

Example. Let us consider the D_4 Argyres-Douglas theory. A BPS quiver for the theory is simply given by an orientation of the D_4 Dynkin diagram

$$\begin{tikzcd} & & \circ_3 \\ \circ_1 \arrow[r] & \circ_2 \arrow[ur] \arrow[dr] & \\ & & \circ_4 \end{tikzcd} \qquad B = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

The Smith normal form of B is simply $\mathrm{diag}(1, 1, 0, 0)$, so we immediately see that this theory has no one form symmetries.

Alternatively, we could engineer the theory from type IIB by using the hypersurface specified by the vanishing locus of

$$W(x, y, z, u) = x^3 + xy^2 + z^2 + u^2. \tag{6.39}$$

Using Orlik's algorithm on this singularity gives perfect agreement with the Smith normal form calculation.

6.3.3 Freed-Moore-Segal fluxes

The defect group captures the charges of possible defects of a theory we wish to engineer from string theory. However, it is not typically the case that the defect group is actually the group of higher form symmetries of the theory. Indeed, this is expected to be the case since the defect group construction neglects the possibility of non-commuting fluxes. Such fluxes are known to be present whenever the internal geometry has torsional cycles [142,143]. To obtain the true higher form symmetry groups from the defect group, one must therefore understand the algebra of non-commuting fluxes.

Consider type IIB on a background of the form $\mathcal{M}_4 \times \mathcal{Y}_6$ where, as before, $\mathcal{Y}_6$ is a Calabi-Yau cone over a manifold Y_5 and $\mathcal{M}_4$ is a closed spin manifold. By performing Hamiltonian quantisation of the theory, and by viewing the transverse direction of the cone as time, we can assign to each spatial slice $\mathcal{N}_9 \sim \mathcal{M}_4 \times Y_5$ a Hilbert space $\mathcal{H}(\mathcal{N}_9)$. This Hilbert space will represent the possible field configurations on such a spatial slice.

Since we have a non-compact internal geometry, one must also specify boundary conditions for fields and fluxes at infinity. In our set up, this corresponds to setting boundary conditions as the 'radius' of the cone tends to infinity. Specifying such a boundary condition is equivalent to picking a state in the boundary Hilbert space $\mathcal{H}(\mathcal{N}_9)$ which describes the appropriate field behaviour. Classically, one would expect all fluxes to commute so we can simultaneously measure them and therefore specify a boundary state in $\mathcal{H}(\mathcal{N}_9)$ to fix all fluxes at infinity. However, it was shown in [142,143], that this is not always the case.

In type IIB, the fluxes are classified by the K-theory group $K^1(\mathcal{N}_9)$. If $\{\sigma_i\}$ generate $\operatorname{Tor} K^1(\mathcal{N}_9)$, then the operators Φ_σ, which measure the flux through a torsional cycle Poincaré dual to σ, form a Heisenberg algebra

$$\Phi_{\sigma_i}\Phi_{\sigma_j} = e^{2\pi \mathrm{i}\ell(\sigma_i,\sigma_j)}\Phi_{\sigma_j}\Phi_{\sigma_i}, \tag{6.40}$$

where $\ell(\cdot,\cdot)$ is the link pairing of cycles dual to the σ_i. As such, we cannot simultaneously measure the eigenvalues of all the flux operators. Instead, we can choose a maximally commuting subset of such operators in order to fix these fluxes at infinity. This can be achieved by picking a maximally isotropic subgroup $L \leq \operatorname{Tor} K^1(\mathcal{N}_9)$. Having chosen L, we can then pick a 'zero flux' state $|0,L\rangle$ such that

$$\Phi_\sigma|0,L\rangle = |0,L\rangle, \tag{6.41}$$

for any $\sigma \in L$. For $\varsigma \in \operatorname{Tor} K^1(\mathcal{N}_9)/L$, we have that

$$\Phi_\varsigma|0,L\rangle = |\varsigma,L\rangle. \tag{6.42}$$

As such, we obtain a basis for the geometric engineering Hilbert space parametrised by $\operatorname{Tor} K^1(\mathcal{N}_9)/L$. Choosing background fluxes for a higher form symmetry can then be done by specifying a state in this basis.

The above discussion eludes to the fact that geometric engineering, a priori, does not necessarily give a single theory, but instead a set of possible theories which differ by choices of mutually commuting fluxes. By specifying these, we end up with a definitive theory whose defects with charges in L are no longer genuine. Therefore, this breaks the defect group to a subgroup which represents the true higher form symmetry group of the theory.

6.4 Decomposition

So far we have discussed how to engineer defects charged under a higher form symmetry and calculate the defect group through quiver computations. However, we have not discussed any possible physical consequences of these symmetries yet. As such, we will briefly discuss an extreme consequence of the presence of certain higher form symmetries.

Decomposition is a phenomenon that occurs in systems with a $(d-1)$-form symmetry where a theory decomposes into the disjoint union of other theories [144–147]. By this we mean that the theory possesses a set of projection operators $\{\Pi_i\}$ such that

$$[\Pi_i, \mathcal{O}] = 0, \tag{6.43}$$

for any operator $\mathcal{O}$ and that

$$\Pi_i \Pi_j = \delta_{ij} \Pi_j, \quad \sum_i \Pi_i = \mathrm{id}. \tag{6.44}$$

The existence of such operators then tells us that the Fock space can be simultaneously diagonalised into eigenmodes of the projectors. Physically, this means that we have several non-interacting theories which forms the overall theory.

A stark consequence of decomposition is the fact that physical quantities should respect the decomposition into theories. The typical example of this is given by the partition function. As first seen in [144], the partition function must be of the form

$$\mathcal{Z}(\mathscr{T}) = \sum_i \mathcal{Z}(\mathscr{T}_i), \tag{6.45}$$

where $\mathscr{T}_i$ are the constituent theories that $\mathscr{T}$ decomposes into. This effect is often called *summing over universes.* While the partition function factorisation in eq. (6.45) is necessary, note that it is not sufficient. Indeed, one must have that *all* physical quantities obey such a decomposition and that the projectors can be built.

The prototypical example of decomposition, and the focus of paper III, is given by σ-models on ineffective orbifolds. An ineffective orbifold $[X/G]$ is simply an orbifold where G acts unfaithfully on X. Mathematically, this isn't really an orbifold in the traditional sense but instead a more sophisticated object called a ***quotient stack*** [148–151]. The technicalities of stacks will not be important for our discussion, but it is important to note that σ-models with target quotient stacks were first considered in [150] and re-examined in [144] where it was pointed out that these theories, naïvely, violate cluster decomposition. Cluster decomposition states that two experiments sufficiently far apart cannot have an effect on each other's outcomes. The σ-models on quotient stacks violate this due to the presence of several non-trivial dimension zero operators.

However, there is a caveat that could resolve the consistency of such models. If the theory decomposes into several non-interacting theories and the extra dimension zero operators belong to different theories, then cluster decomposition is not violated. This is indeed the case, as it was shown in [144] that if the trivially acting part of G is central $N \leq Z(G)$ then the QFT associated to $[X/G]$ has a one form symmetry given by N and decomposes as

$$\mathbf{QFT}([X/G]) = \mathbf{QFT}\left(\bigsqcup_{\rho \in \hat{N}} \left[\frac{X}{(G/N)} \right]_{\hat{\omega}(\rho)} \right), \tag{6.46}$$

where $\hat{\omega}(\rho)$ denotes the discrete torsion of the theory [152–156]. In particular, it is given by the image of the extension class $\rho \in \mathrm{H}^2(G/N, N)$ of the extension

$$1 \to N \to G \to G/N \to 1. \tag{6.47}$$

An explicit example of this decomposition is shown in section 2 of paper III where the torus partition function exhibits the decomposition nicely.

Finally, let us comment on why one would expect decomposition when there is a $(d-1)$-form symmetry. The charged objects of such a symmetry are codimension-1– that is, they are *domain walls*. However, not all domain walls are associated to a symmetry. Indeed, the typical domain wall can have a finite mass in a compact subregion of it. This allows for possible tunnelling between vacua. On the other hand, if a domain wall is associated to a symmetry then the mass must be infinite and no tunnelling can occur. This distinguishes decomposition from super-selection sectors, for example, as the tunnelling can occur and no $(d-1)$-form symmetry is present. It was recently shown in [157], that gauging such a symmetry simply projects onto one of the constituent theories labelled by a dual (-1)-form symmetry.

Introduction to attached papers

Paper I

The first attached paper deals with two classes of four dimensional $\mathcal{N} = 2$ theories and their one-form symmetries. The first set of theories are the so-called $G^{(b)}[k]$ theories which correspond to isolated singularities defined by the hypersurface equation

$$f(x,y,u) + zg(x,y,u,z) = 0. \tag{6.48}$$

Such singularities are called compound du Val and are typically non-isolated. In [158], the isolated compound du Val singularities were classified to give the $G^{(b)}[k]$ theories. The second class of theories are defined by the same hypersurface equation as above, but with the caveat that z should be viewed as a $\mathbb{C}^{\times}$ variable instead. This engineers the $D_k^b(G)$ theories instead.

Given the similarities in their construction, it is no surprise that there is a relation between the two. Starting with a given $D_k^b(G)$ theory, one can trigger a Maruyoshi-Song RG flow to end up at the $G^{(b)}[k]$ theory in the IR [159]. This raises the natural question of 'how are the higher-form symmetries of these two sets of theories related'?

The paper proceeds as follows. First we analyse the case of $b = h^{\vee}(G)$, the dual Coxeter number of G, which have well known BPS quivers [160,161]. From this we apply the methods mentioned in section 6.3 to extract the one-form symmetries of the theories. We find that there is total agreement between the one-form symmetries of the two, suggesting that the Maruyoshi-Song flow of this type preserves such symmetries. We then extend this analysis to the $b \neq h^{\vee}(G)$ by first using Orlik's algorithm to calculate the defect groups for the $G^{(b)}[k]$ theories. To cross check this result, we then calculate the BPS quiver for a subclass of these theories through the methods mentioned in section 4.3 and recover the same answer. Finally, we conjecture that in general the defect groups of the $D_k^b(G)$ theories are the same as that of the $G^{(b)}[k]$ theories for any value of b and k.

Paper II

In section 5.3 we saw that there was a generalisation of the McKay correspondence to subgroups of SU(3). While the orbit spaces $X/\sim$ have generically non-isolated singularities, one can still consider M-theory on such a geometry and ask what physical information one can understand from the geometry. One such piece of information that we can get is the 5d BPS quiver. That is, the BPS quiver of the theory compactified further on a circle. From this it is easy to obtain the defect group of such theories. The main purpose of this theory is to classify the higher symmetries of 5d orbifold theories through both quiver and non-quiver means.

We first start by outlining an algebraic topology approach which allows us to find the fundamental group of the orbifold explicitly. Taking the abelianisation then gives us the desired defect group. This gives us a closed form answer for the defect group, but this is difficult to calculate in examples. We then use the McKay correspondence to form the BPS quivers for the theories, giving concrete answers for any such orbifold.

With the one-form symmetries classified, we then discuss the possibility of there being a 2-group symmetry. We posit that there is a short exact sequence that dictates whether there is a 2-group or not.

Paper III

Paper III continues the study of McKay quivers initiated in paper II, albeit from a different point of view.

When considering the McKay quivers for a given group G, it is easy to choose representations of G for which its McKay quiver is disconnected. This occurs whenever the representation is unfaithful. While this may seems physically unintuitive, since the orbifold group would not act faithfully, it is still interesting to ask if one can describe the resulting quiver components in terms of orbifolds and if there is a physical counterpart to this description. The main aim of this paper is to motivate the idea that the decomposition of orbifold σ-models outlined in section 6.4 dictates the components of a McKay quiver.

Within the paper we present several examples where this equivalence is evident and discuss the quivers for groups defined through a central extension of the image of a given representation. In this discussion we restrict to a purely representation and character theoretic methodology thereby deriving formulae for these quivers compatible with the decomposition formulae.

Paper IV

As mentioned in section 3.5, the stratification of the Coulomb branch gives us great insight to the possible scale invariant geometries required of superconformal field theories. However, even at rank-2, there are myriad possibilities for the stratification which leaves a full classification difficult. The aim of this paper is to define a quantity which can single out a tractable subset of theories which can be classified. We call this quantity the characteristic dimension of the theory.

The characteristic dimension can only take values in the set of allowable rank-1 scaling dimensions. Furthermore, if it is not equal to 1 or 2 then the Coulomb branch geometry is said to be *isotrivial.* Mathematically, this means that the fibration defining the Donagi-Witten integrable system has fibres that are isomorphic as abelian varieties at generic points of the Coulomb branch. Physically, it means that the holomorphic couplings of the theory are locally constant. Equivalently, this means that the theories support on the singular strata are genuine SCFTs and not IR free theories. As an example, all $\mathcal{N} \geq 3$ theories are immediately isotrivial.

The paper is organised as follows. We start by motivating the characteristic dimension through considerations of R-symmetry and monodromies before moving on to some specifics about the interplay between the characterstic dimension and the Donagi-Witten integrable system. We then round out the paper by predicting the existence of two $\mathcal{N} = 3$ theories constructed from complex crystallographic reflection groups.

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Svensk Sammanfattning

Kvantfältsteori utgör för närvarande vår bästa beskrivning av naturen. Under det senaste århundradet har kvantfältsteorin reproducerat många experimentella resultat till en häpnadsväckande grad av noggrannhet samt kunnat förutsäga existensen av nya partiklar. Trots detta lider den av ett flertal matematiska problem. Ett sådant problem är att det ofta bara går att erhålla approximativa resultat via störningsräkning i en svagt kopplad regim, och inte exakta resultat. Om man begränsar systemet genom att införa supersymmetri blir emellertid exakta beräkningar möjliga. I denna avhandling utforskar vi diverse exakta egenskaper hos kvantfältsteori genom att arbeta med teorier med åtta superladdningar.

Mycket av materialet i avhandlingen syftar till att förstå kvantfältsteorier genom deras BPS-koger — grafiska objekt som som beskriver världslinjerna till BPS-tillstånd. Vi använder den geometriska konstruktionen av dessa objekt för att bestämma de högre symmetrierna i många teorier. Vi relaterar också frånkopplade koger i orbifoldfallet till uppdelningen av tvådimensionella sigma-modeller..

Ett andra problem som angrips i avhandligen är klassificeringen av fyrdimensionella konforma teorier med åtta superladdningar. Mer specifikt definierar vi en storhet som låter oss identifiera teorier vilkas integrabla Donagi–Witten-system förenklas enormt. Förenklingarna gör klassificeringsproblemet mer hanterligt för dessa teorier, och vi påbörjar studiet av dem genom att presentera två nya förmodade teorier av rang 2.

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